# The National Value of Local Outdoor Recreation over the Past 20 Years

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**Abstract**

I estimate the value of local outdoor recreation in the United States from 2003 to 2023 using the American Time Use Survey and a travel-cost model. I track changes in value over time, which informs if the places that enable local recreation have been managed sustainably, and across demographic groups, which aids in assessing if these places are equitably distributed. I compare two approaches for pricing travel time: (i) welfare-based willingness to pay, valuing time at one-third of own wage, and (ii) accounting values using a replacement wage consistent with household production accounts. Average value per local recreation trip, which requires less than 30-minutes of travel time, is stable at $16–$18 (2023 USD), with temporary increases around the Great Recession and the COVID-19 pandemic. Scaling by trips and population, the national annual value is $231–$241 billion in 2023, roughly $100 billion higher than in the early 2000s. While aggregate trends are similar under both pricing approaches, distributional conclusions differ substantially: using a replacement wage compresses the richest quintile's per-trip value from about 10 times larger than the poorest to only 1.5 times larger. I also use the framework from Drupp et al. (2024, 2025) to infer changes in the natural capital stock that enables recreation. Combining value-per-trip trends with consumption growth, I find that recreation-enabling assets have been roughly maintained over the past two decades, consistent with sustainable management at the national level. This paper produces a scalable, repeatable measure of local recreation benefits at the national level, connects non-market valuation with accounting-consistent pricing, and provides a practical tool for tracking growth of an ecosystem service's enabling natural capital stock when direct measurement is infeasible.

## 1 Introduction

Evaluating progress toward sustainable development requires measuring changes in welfare or real wealth through time, but traditional economic statistics often omit ecosystem services that contribute to welfare and natural capital assets that contribute to wealth (Nordhaus and Tobin, 1973; Arrow et al., 2004; Dasgupta and HM Treasury, 2021). Valuing environmental benefits is necessary when natural resources have competing market and non-market uses and society seeks to achieve sustainable management such that current generations meet their needs without compromising the ability of future generations to do the same (World Commission on Environment and Development, 1987; Solow, 1991). Recreation opportunities are a

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major ecosystem service provided by nations' land and water bodies. The benefits of outdoor recreation are central to policy appraisals such as benefit cost analyses about the non-market value of natural spaces. In addition to needing to estimate the non-market value of outdoor recreation over time to assess sustainability of management, there is a gap in understanding who benefits from these services (Drupp et al., 2025a). It is important to understand the distribution in benefits across the current generation, as well as across time, in order to manage resources in a way that improves well-being. Indeed, Robert Solow (1991) remarked "there is something faintly phony about deep concern for the future combined with callousness about the state of the world today."

This paper estimates the value of local outdoor recreation in the United States, how the value has changed over the last 20 years, and how it varies across the income distribution and racial groups. I evaluate how the estimated values depend on how time is priced in travel cost models. To do so, I first estimate annual willingness to pay (WTP) for local outdoor recreation from 2003 to 2023 using one-third own wage as the opportunity cost of time, and characterize the distribution of benefits across income quintiles and racial groups. I then contrast the WTP estimated (an opportunity-cost-of-time approach) to the value estimated when I use a replacement wage to convert time to a monetary unit (the methodology used in household production accounts). This comparison is conceptually important because the opportunity-cost approach ties estimated values to ability to pay (*i.e.,* monetary income). However, local outdoor recreation is a public ecosystem service typically provisioned for free and, thus, consuming the service is not constrained by an individual's monetary income. Therefore, the replacement wage approach provides a way to monetarily value a free and public ecosystem service with a method that is methodologically consistent with other national statistics, but is not functionally dependent on the individual's income.

Beyond estimating value of a local recreation trip with different travel cost methods, this paper leverages the temporal dimension of my value estimates to assess changes in the underlying natural capital stock that enables outdoor recreation. Using the framework developed by Drupp et al. (2024, 2025b), which adjusts the value of ecosystem services using relative price changes caused by national income growth, I infer whether recreation-enabling assets are being maintained or depleted over the 20-year study period. This framework allows me to answer whether the enabling assets for recreation have been managed sustainably over my study period without having to observe these assets directly.

I document several stylized facts about local outdoor recreation in the United States. Using the American Time Use Survey, I show that most outdoor time is spent doing everyday activities such as walking, caring for pets or children, and socializing. Most outdoor recreation trips are local, requiring less than 30 minutes of one-way travel. The median amount of time spent recreating (dwell time) is 30 minutes, and 90 percent of recreation trips' dwell time is 2 hours or less. Participation rates are high across income quintiles and racial groups. In my empirical analyses, I find that the WTP per local recreation trip is $18 (2023 USD) when using one-third own wage, and the accounting value of a trip is $16 when using a replacement wage. Following, the welfare provided by local recreation in the United States in 2022 is $241 billion and the accounting value is $231 billion. Both the national welfare estimate and accounting value have increased about $100 billion since 2004. This increase is driven by recreators taking more trips and by population growth, illustrating the nonrival nature of local outdoor recreation: value per trip has remained stable even as more people use these resources.

How time is priced matters primarily for distributional conclusions. Using a replacement wage rather than one-third of own wage has little effect on the aggregate time trend in the value of local recreation, but it substantially compresses income gradients. Under the opportunity-cost approach, the richest quintile's

value per trip is about 10 times greater than the poorest quintile's; under the replacement-wage approach, this ratio shrinks to about 1.5 times. Conditional on income, the ordering of racial groups' per trip value is unchanged, and the estimates do not vary significantly. Because travel costs for local trips are almost entirely composed of travel time, this comparison isolates how the modeler's choice of time valuation affects estimates of the distribution of benefits across income.

Applying the framework from Drupp et al. (2024, 2025b) to infer changes in the natural capital stock enabling recreation, I find that recreation-enabling assets have been roughly maintained over the past two decades. By combining estimated trends in value per trip with consumption growth and established income elasticities, I calculate the implied growth rate of the enabling asset stock. Despite substantial period-to-period volatility, periods of apparent expansion offset periods of contraction, yielding near-zero average growth. This finding suggests that, in aggregate, the infrastructure and natural areas enabling local recreation have kept pace with growing national demand, which is consistent with sustainable management at the national level. This methodology is particularly valuable for local outdoor recreation because the enabling assets (*e.g.,* parks, greenways, tree-lined streets, sidewalks, schoolyards, and churchyards) are dispersed and heterogeneous, making direct observation difficult. Existing U.S. land accounts do not include recreational land, partially because it is difficult to classify Wentland et al. (2020). Rather than attempting to inventory this diffuse stock, the framework allows me to infer whether recreation-enabling assets are growing or shrinking based on observed changes in willingness to pay.

Using the time people spend traveling to recreate as the travel cost has become a prolific way to estimate the willingness to pay for non-market environmental benefits. Travel cost models are the dominant method for valuing outdoor recreation (Haab and McConnell, 2002b; Freeman et al., 2014; Parsons, 2017; Phaneuf and Requate, 2017). Travel cost models have been used to value improvements in water quality (Keiser, 2019; Griffiths et al., 2012; Egan et al., 2009; Bockstael et al., 1987), the costs of oil spills (English et al., 2018; Hausman et al., 1995), the costs of infectious disease exposure (Day, 2020; Berry et al., 2018), the benefits of agricultural practices (Hansen, 2007; Sullivan et al., 2004; Hansen et al., 1999), the benefits of species conservation (Gürlük and Rehber, 2008; Loomis et al., 2000), and the costs of endangered species protection (Dundas et al., 2018). Given travel cost models' broad application to diverse environmental assets and ecosystem services beyond outdoor recreation, understanding how modeling assumptions affect estimates and developing methods for repeated estimation are crucial for consistent, comparable valuation across contexts.

The first contribution of this paper is demonstrating the ability to repeatedly estimate the value of local recreation at a national scale, filling a current research gap (Fenichel, 2024). Despite more than 50 years of innovation in travel cost methods and their influence on management decisions through benefit cost analyses and litigation (Parsons, 2017), historical data constraints have limited travel cost models to estimates for a site or region and for the average person at a single point in time (Lupi et al., 2020). A repeated national-scale estimate is needed to assess whether local recreation opportunities are being managed sustainably. While the U.S. Bureau of Economic Analysis's Outdoor Recreation Satellite Account (ORSA) tracks spending on recreation's complementary market goods through time (Highfill et al., 2018), such trends may reflect broader consumption rather than environmental management. The ORSA also does not consider market expenditure on recreation trips that occur within 50 miles of home, thereby completely excluding the value of local recreation.

The second contribution is describing how the value of local outdoor recreation varies across the income distribution and racial groups, and demonstrating how these distributional patterns depend on how modelers

price travel time. Until recently, the intragenerational distribution of environmental benefits received limited attention and is less understood than the distribution of economic resources (*i.e.,* income, wealth) (Cain et al., 2024; Drupp et al., 2025a; Fenichel, 2019). I compare and contrast a traditional opportunity cost of time approach (typical for welfare estimates in a recreation demand setting) to the use of a replacement wage (typical in national accounting), showing that methodological choices have substantial implications for equity conclusions. Because the use of replacement wages may be unfamiliar to some readers, I provide a background in Section 2 on their use in household production accounting.

The third contribution is demonstrating how repeated WTP estimates can reveal changes in recreation-enabling natural capital when direct measurement of the stock is infeasible. Local recreation presents a unique challenge for natural capital accounting because the enabling infrastructure (*e.g.,* parks, greenways, tree-lined streets, schoolyards, and other mixed-use spaces) is too dispersed and heterogeneous for cost-effective direct observation. Using the framework developed by Drupp et al. (2024, 2025b), I combine my estimated WTP trends with established income elasticities to recover the implicit growth rate of the recreation-enabling asset stock. This demonstrates a practical approach for integrating revealed preference estimates into sustainability assessments of how a natural capital asset is being managed when traditional stock inventories are difficult to directly observe.

## 2 Background: Replacement Wages, Household Production, and Time Valuation

This section discusses the role of replacement wages in valuing household production, and the role they can play in harmonizing recreation demand modeling with existing national accounting methodologies. First, it is useful to understand the difference between two accounting boundaries that are relevant for tracking welfare through time: the System of National Accounts (SNA) boundary and the household production boundary (Office of Science and Technology Policy et al., 2023). The SNA boundary focuses on monetary flow information from market exchanges and underpins traditional economic statistics such as gross domestic product (GDP) and inflation. The household production boundary captures in-kind income from services that individuals and households produce for themselves, which are generally beyond the SNA production boundary but nonetheless important to economic prosperity (Office of Science and Technology Policy et al., 2023). The U.S. National Strategy to Develop Statistics for Environmental-Economic Decisions explicitly identifies outdoor recreation as a household-produced service that falls within both these boundaries. Measuring the value of household production is an important step towards empirically measuring a national welfare statistic that captures well-being from both market consumption and self-produced services. Measuring such a welfare statistic requires considering an accounting boundary that is the union of the SNA and the household production accounting boundaries (Weitzman, 1976; Sefton and Weale, 2006; Jorgenson and Slesnick, 2014; Fenichel, 2024).

The value of outdoor recreation should, theoretically, be measured under both the SNA and household production accounting boundaries because recreation drives market transactions and also requires people produce the service of recreation for themselves via their own travel time (Office of Science and Technology Policy et al., 2023). Currently, outdoor recreation already generates market transactions (*e.g.,* fuel, lodging, equipment) that are recorded in GDP. The Bureau of Economic Analysis's Outdoor Recreation Satellite Account (ORSA) reclassifies these transactions to measure the "outdoor recreation economy" (Headwaters Economics, 2021; Highfill et al., 2018). To be clear, the ORSA is a reclassification of values already included

in GDP, it is not a measurement of a value that is "missing" from GDP.

The BEA also maintains a satellite account that measures household production. National accountants price non-market production that has "near market" analogs. To determine if an activity is a near market service, the BEA asks "Would someone pay another person (a 'third person' from outside the home) to perform the activity?" (Bureau of Economic Analysis, 2018b). If yes, then that activity is considered near-market. Current near market services included are activities such as cooking, cleaning, or simple repairs. National accountants use replacement wages to value these services. In theory, the replacement wage is the wage a market producer (rather than household) would earn for providing the same service, adjusted for differences in quality between home and specialist production (National Research Council et al., 2005; Landefeld et al., 2009). In practice, the BEA uses the average wage rate of a general-purpose domestic homeworker as the replacement wage for all tasks they price in the Household Production Satellite Accounts (Bureau of Economic Analysis, 2018b). The BEA's Household Production Satellite Account observes the time U.S. residents spend producing goods and services at home using the American Time Use Survey (ATUS) (Landefeld et al., 2009; Bureau of Economic Analysis, 2018a). The value of outdoor recreation is not currently included in the U.S. household production accounts despite it being a service that individuals produce for themselves using their travel time.

Using a replacement wage to value time is conceptually different from using the opportunity cost of time, which is typically used in recreation demand models to estimate WTP measures:

1. **Welfare (WTP) via opportunity cost of time.** In a welfare framework, travel-time costs are converted to willingness to pay using the marginal opportunity cost of time, *i.e.,* the marginal after-tax wage at which an individual can sell an additional unit of time (Nordhaus, 2006). This is a counterfactual valuation that asks what the person could have earned if they had supplied the same time to the market, presumably in their primary job. Historically, limited flexibility in hourly work for large parts of the population (*e.g.,* salaried workers, retirees) led economists to proxy the opportunity cost of time with a fraction of the average wage. The conceptual target is the marginal wage, not the average.

2. **Accounting value via replacement wage.** In household production accounting, time is priced by the market value of the service produced. For near-market services, accountants ask: what would an individual earn if they sold this service or good on the market (rather than producing and consuming it themselves)? This is a realized-service valuation that makes non-market production commensurate with market production by focusing on what was produced, rather than what could have been produced with the same amount of time (National Research Council et al., 2005; Landefeld et al., 2009; Bureau of Economic Analysis, 2018b).

These two approaches coincide if and only if the individual's marginal opportunity cost equals the relevant replacement wage. They would not typically be equal because the opportunity cost depends on the person's marginal earning possibilities, whereas the replacement wage depends on the market price of the service produced. Therefore, welfare-based WTP and accounting values that monetize the same service may diverge even when they use the same underlying time inputs.

Applying this logic to recreation, it is useful to distinguish travel time from on-site leisure time. Travel time is a near market service with close market analogs and can be priced using a replacement wage consistent with household production accounting. Market analogs include professional driving such as taxis, and related transportation services such as driving for Uber. By contrast, on-site leisure is a "personal" good or service

that must be produced by the individual. Personal goods and services have no near market equivalent (an individual would not pay someone else to leisure for them). Therefore, personal goods and services are not included in household production accounts (Nordhaus, 2006). Standard travel-cost models therefore yield a lower bound on the benefits of outdoor leisure: individuals would not incur travel costs to produce the trip if the value of the on-site benefits were not at least as large.

Finally, connecting travel-cost valuation to household production accounting aligns empirical practice with the broader objective of building welfare-relevant national statistics that integrate market and non-market production. In the next section, I formalize these ideas by adapting a Becker household production framework to show how outdoor recreation can be considered household production, and how either the opportunity cost of time or replacement wages can be used for valuing travel time.

# 3 Conceptual Framework: Adapting the Becker Household Production Model

The Becker (1965) Household Production Model considers an individual who maximizes her utility by consuming commodity goods that are comprised of various market and non-market goods or services. Outdoor recreation trips are an example of such a commodity good because they can consist of market goods (*e.g.*, hiking boots, picnic blankets) and non-market goods (*e.g.*, travel time). The Becker Model conceptualizes an individual, or household, as both a consumer and producer because the consumer produces the commodity good for herself. The commodity good is written as

$$Z_i = f_i(\mathbf{x}, \mathbf{T})$$

where $\mathbf{x}$ is a vector of market goods required to produce the commodity good $Z_i$, $\mathbf{T}$ is a vector of the time inputs, and $f_i(\cdot)$ is the production function for good $Z_i$.

The consumer's utility function can be written as

$$U(Z_1, \ldots, Z_n) \equiv U(f_1, \ldots, f_n) \equiv U(x_1, \ldots, x_m; T_1, \ldots, T_m).$$

Maximizing utility is constrained by a full resource constraint that considers money spent on required input bundles of market goods $\mathbf{x}$ and time required to make the good or service $\mathbf{T}$. This resource constraint is commonly referred to as full income (Haab and McConnell, 2002a). I assume maximizing utility requires allocating the full income to the consumption of commodity goods and services. When assuming that the opportunity cost of time is the average wage rate $w$, full income ($S$) can be written as

$$\sum_i \left(\mathbf{p}_i \mathbf{x}_i + w\mathbf{T}_i\right) Z_i = S^{\text{hourly wage}} \tag{1}$$

where $\mathbf{p}_i$ is a vector of prices for the vector of market goods $\mathbf{x}_i$ required to produce the $i$th commodity good $Z_i$ and $\mathbf{T}_i$ is a vector measuring time spent doing various non-market activities to produce $Z_i$. In this case of using the average hourly wage to calculate full income, the full price of the commodity good $Z_i$ is

$$\pi_i^{\text{hourly wage}} = \mathbf{p}_i \mathbf{x}_i + w\mathbf{T}_i. \tag{2}$$

Becker (1965) recognizes that "marginal, not average, prices are relevant for behavior" (pg. 499). If an

individual cannot earn her average wage beyond a set number of hours, as is true for salaried workers or hourly workers with inflexible hours, then Equation 1 overstates her full income. Equation 1 is also not applicable for retirees or children, who may have income and time to spend on goods but not an average hourly wage. Therefore, he presents a general case of full income,

$$\sum_i \mathbf{p}_i \mathbf{x}_i Z_i + L(T_1, \ldots, T_n) = S^{\text{loss func.}}, \tag{3}$$

where $L(\cdot)$ is a "loss" function that measures the income that an individual forgoes by taking the time to produce commodity goods $\mathbf{Z}$ for herself rather than selling that time on the market for the service produced. In this generalized case that monetizes required household production time with a loss function that measures what the same activities would earn on the market (rather than using the individual's average wage rate as was the case in Equation 2), the price of the commodity good $Z_i$ is

$$\pi_i^{\text{loss func.}} = \mathbf{p}_i \mathbf{x}_i + L_i(\mathbf{T}_i). \tag{4}$$

It is worth noting that market goods also require travel time that is currently missing from the BEA's Household Production Satellite Account. Both of the prices I have so far defined ($\pi_i^{\text{hourly wage}}$, $\pi_i^{\text{loss func.}}$) incorporate travel time $\mathbf{T}_i$ alongside required market inputs $\mathbf{p}_i \mathbf{x}_i$. For example, the "full price" of dining at a restaurant includes not only the bill paid at the establishment, but also the household production time spent driving there and back. This travel time represents a near-market service that individuals could sell on the market as a driver for hire, but instead is time spent driving to dine at a restaurant. This travel time also remains outside current household production accounts. Outdoor recreation is not unique in this regard; rather, it is one of many activities, both market and non-market, whose full cost includes travel time that national accounts do not yet capture.

In the empirical sections of this paper, I estimate welfare-based willingness to pay (WTP) for a local outdoor recreation trip using the conventional opportunity-cost-of-time approach, pricing travel time at a fraction of the individual's average wage ($\alpha w$), with $\alpha = 1/3$ as in standard practice (Cesario, 1976; Lupi et al., 2020). Second, I estimate an accounting value by pricing the service actually undertaken (travel) at a replacement wage $r(k)$ where $k$ is the mode of travel, consistent with household production accounting (Bureau of Economic Analysis, 2018b; Landefeld et al., 2009). The former is a counterfactual, welfare interpretation ("what the person could have earned"); the latter is a realized-service, accounting interpretation ("what the produced service is worth on the market"). The two coincide only if $\alpha w = r(k)$, which likely does not hold in general.

To find the travel cost of a recreation trip using the methodology from household production accounting, I parameterize the income loss function $L_i(\mathbf{T}_i)$ from Equation 4 using a replacement wage. In the case of an outdoor recreation trip, the vector $\mathbf{T}_i$ is composed of the travel time $t_i^{\text{near}}$ (the near-market service) and on-site time $t_i^{\text{pers}}$ (the personal service). The loss function can then be written as

$$L_i = r(k)\, t_i^{\text{near}} + \lambda\, t_i^{\text{pers}} \tag{5}$$

where $r(k)$ is the replacement wage that could be earned if the individual chose to sell her travel time on the market (the rate depends on the travel mode $k$: driving, biking, walking, public transport, etc.). The parameter $\lambda$ is the wage that the individual could earn if she sold her on-site leisure time on the market, which I assume to be zero because leisure is a personal good and thus has no near-market analog in household

production accounts. Plugging Equation 5 into Equation 4, the accounting price of an outdoor recreation trip (i.e., the travel cost used for the accounting value) is

$$\pi_i^{\text{acc.}} = \mathbf{p}_i \mathbf{x}_i + r(k)\, t_i^{\text{near}}. \tag{6}$$

For the welfare (measured as WTP) approach commonly used in travel-cost models, I price travel time at a fraction of the individual's average wage as

$$\pi_i^{\text{WTP}} = \mathbf{p}_i \mathbf{x}_i + \alpha\, w\, t_i^{\text{near}}, \tag{7}$$

where $w$ is the average wage rate and $\alpha$ is set to one-third based on revealed- and stated-preference evidence about time valuation, as well as precedent in travel cost models (Cesario, 1976; Lupi et al., 2020). The fraction of the wage rate, $\alpha w$, is intended to approximate the marginal opportunity cost of time. In my empirical analysis, I report WTP estimates based on $\pi_i^{\text{WTP}}$ and contrast them with accounting values based on $\pi_i^{\text{acc.}}$, comparing both aggregate time trends and distributions across income and racial groups.

Estimating the value of outdoor recreation using both pricing approaches allows me to test the sensitivity of my conclusions along two dimensions. First, I assess temporal sensitivity: whether the observed trends in recreation values from 2003 to 2023 depend on the choice of time valuation method. If aggregate welfare trends are similar under both approaches, this suggests that conclusions about changes in natural capital are robust to pricing methodology. Second, I evaluate intratemporal distributional sensitivity: whether conclusions about how recreation values vary across income quintiles and racial groups depend on time pricing. Because travel costs for local trips are almost entirely composed of time rather than market goods, this comparison isolates how modeling choices regarding the pricing of travel affect equity assessments, independent of other methodological decisions.

# 4 Data and Motivating Facts

My analyses use the American Time Use Survey (Flood et al., 2024), which the U.S. Bureau of Labor Statistics conducts. It is the most comprehensive survey of its kind in the United States and has been used by economists to understand patterns and behaviors in daily life (Burda et al., 2013; Aguiar et al., 2013; Bayham et al., 2015; Berry et al., 2018; Cubas et al., 2021; Bayham et al., 2021; Chan and Wichman, 2022).

Individuals are asked to report their time spent on all activities in a 24-hour period. Respondents report their primary activities, location, duration, and whether the activity was done with others. There are 17 major activity categories and over 400 six-digit coded activities (U.S. Bureau of Labor Statistics, 2024). The ATUS is a stratified survey, and weekends are oversampled to capture a greater variety of behavior. All statistics I present account for the respondents' sample weight. Respondents are linked to the Current Population Survey (CPS) to observe key demographic information including county of residence, income, and other demographic characteristics.

I identify outdoor leisure activities using two filters. First, I identify activities that occurred "Outdoors - not at home" using the location codes provided in the ATUS. Leisure activities outside at home (*i.e.*, in an individual's yard) are not included in my definition of local outdoor leisure. Second, I filter to 99 six-digit leisure activities that occur outdoors, all of which are presented in Appendix Section A.1, Table 4. These 99 leisure activities are deemed as outdoor leisure if they are activities someone may participate in with their free time outside of work.

Table 1: **Outdoor Activity Categories:** Summary of 30 outdoor activity categories (aggregated from 99 six-digit ATUS codes), sorted by average annual hours per U.S. resident. Activities are limited to those occurring outdoors (not at home) with round-trip travel times less than 60 minutes. All estimates are weighted by ATUS sample weights for the 2003-2023 period. Note that common outdoor activities like walking dominate, while traditional recreation pursuits (fishing, hiking, climbing) rank lower.

| | Activity Category | Avg. Annual Hours (per U.S. Resident) |
|---|---|---|
| 1 | Walking | 8.80 |
| 2 | Pet Care and Leisure | 7.70 |
| 3 | Socializing and Relaxing | 5.74 |
| 4 | Child Care and Leisure | 3.49 |
| 5 | Running | 2.24 |
| 6 | Fishing | 2.13 |
| 7 | Hunting | 1.59 |
| 8 | Water sports | 1.56 |
| 9 | Biking | 1.39 |
| 10 | Golfing | 1.20 |
| 11 | Hiking | 0.85 |
| 12 | General Sports | 0.84 |
| 13 | Soccer | 0.57 |
| 14 | Baseball | 0.55 |
| 15 | Basketball | 0.53 |
| 16 | Football | 0.37 |
| 17 | Softball | 0.30 |
| 18 | Racquet Sports | 0.30 |
| 19 | Snow Sports | 0.29 |
| 20 | Religious Activity | 0.29 |
| 21 | Rollerblading | 0.18 |
| 22 | Adult Care and Leisure | 0.15 |
| 23 | Equestrian Sports | 0.15 |
| 24 | Volleyball | 0.15 |
| 25 | Vehicle Touring/Racing | 0.11 |
| 26 | Climbing | 0.07 |
| 27 | Rugby | 0.03 |
| 28 | Rodeo Competitions | 0.02 |
| 29 | Hockey | 0.02 |
| 30 | Extracurricular club activities | 0.01 |

In Table 1, I group the 99 six-digit activities into 30 categories and calculate the average hours a U.S. resident spends participating in the activity per year. The most common outdoor leisure activity is walking, followed by Pet Care and Leisure, Socializing and Relaxing, and then Child Care and Leisure. These activities are not stereotypical outdoor recreation activities, like Fishing and Hunting (sixth and seventh most common), Hiking (eleventh), Snow Sports (nineteenth), or Climbing (twenty-sixth).

I identify indoor leisure activities that may be substitutes for outdoor leisure using two filters. First, I filter to activities that occurred at one of the following places: bar or restaurant, other store/mall, school, library, gym/health club, or other places (not at home). Second, I identify 90 six-digit activities that are indoor leisure activities, all of which are listed in Appendix Section A.1, Table 5. Again, leisure activities are those that people participate in using time outside their work hours. I group similar activities into six different substitute activity categories, which are listed in Table 2, along with the average annual time

Table 2: **Indoor Activity Categories:** Summary of six indoor substitute activity categories (aggregated from 90 six-digit ATUS codes), sorted by average annual hours per U.S. resident. Activities occur at indoor locations away from home (bars, restaurants, stores, gyms, etc.). All estimates are weighted by ATUS sample weights for the 2003–2023 period. Americans spend substantially more time on indoor leisure than outdoor leisure, with eating/drinking and shopping dominating.

| | Substitute Activity Category | Avg. Annual Hours (per U.S. Resident) |
|---|---|---|
| 1 | Eating or Drinking | 76.46 |
| 2 | Shopping | 60.44 |
| 3 | Socializing | 32.73 |
| 4 | Entertainment | 30.39 |
| 5 | Playing Sports or Games | 27.40 |
| 6 | Hobby or Relaxing | 11.24 |

Americans spend participating in each. It is worth noting that Americans spend much more time doing indoor leisure as compared to outdoor leisure.

I calculate the travel time for indoor and outdoor leisure activities using individuals' daily diaries. Once an indoor or outdoor activity is identified, I sum the travel time that occurs within two activities before and after the outdoor activity to get the round-trip travel time.

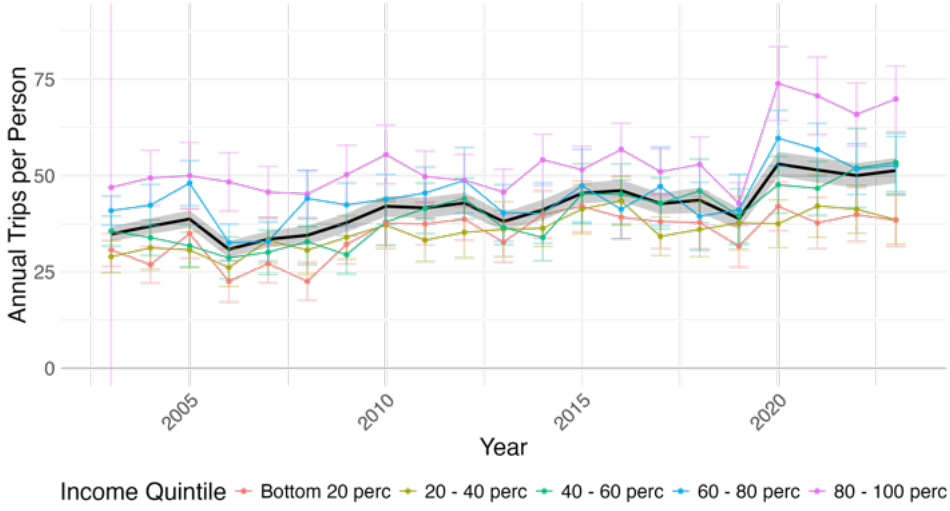
I classify an outdoor leisure activity as local if the activity's total travel time is less than one hour. Seventy-eight percent of outdoor leisure activities require one hour or less of travel time. I chose this as the cut-off for a local trip because the average one-way commute time in metropolitan areas was 26 minutes from 2012 through 2016 (U.S. Census Bureau, 2017), which is the middle of my study period. Additionally, 62 percent of outdoor leisure activities require 50 minutes of travel versus 78 percent requiring one hour. Using a one-hour cut-off, rather than the approximately 50-minute round-trip travel time for the average work commute, allows me to include the 16 percent of recreational trips that require between 50 and 60 minutes of travel in my analyses. This helps provide additional statistical power when subsetting the data by time periods or demographic groups.

The median round-trip travel time for outdoor leisure is 30 minutes. Most outdoor leisure activities require an hour or less of travel time, regardless of income quintile or racial group (Appendix Section A.2, Figure 5). Indoor leisure activities require just over 30 minutes of travel. There are no significant changes in average travel time for outdoor leisure activities across income or racial groups between 2003 and 2023 (Appendix Section A.2, Figure 6).

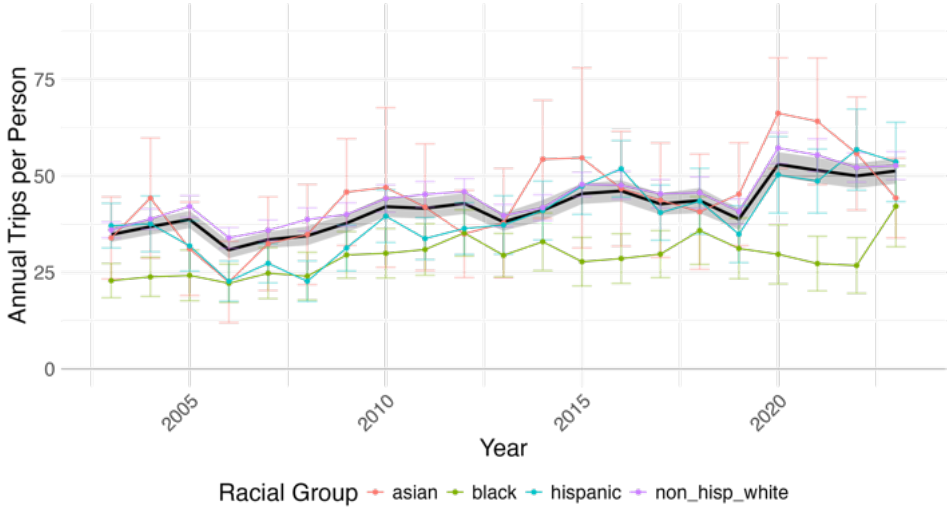
The average number of annual local trips per person has increased from 37 to just over 50 for all U.S. residents, regardless of income or race (Figure 1, see black line). This increase has occurred across all income quintiles (Figure 1a), though the increase is most prominent for the highest income quintiles. Notably, the lowest income quintile saw a significant increase in the average number of annual trips taken between 2008 and 2010, around the time of the U.S. financial crisis. The highest income quintile experienced a similar but larger increase between 2019 and 2020, during the COVID-19 pandemic. In both cases, the increased number of annual trips persists through time. Over the study period, the number of outdoor recreation trips has also increased for all racial groups (Figure 1b).

Figure 1: **Annual Local Recreation Trips per Person, 2003–2023.** Panel (a) shows trends by income quintile. Trip-taking has increased for all quintiles, with notable jumps: the lowest income quintile increased trips around the 2008 financial crisis, while the highest income quintile increased trips during the COVID-19 pandemic. Both increases persisted through subsequent years. Panel (b) shows trends by race/ethnicity. Trip-taking has increased for all racial groups over the study period. In both panels, the black line (with shaded 95% confidence interval) shows the national average across all individuals, regardless of income or race. Estimates are calculated from ATUS data using sample weights.

(a) Annual Local Recreation Trips per Person by Income Quintile




(b) Annual Local Recreation Trips per Person by Race/Ethnicity



# 5 Methods

## 5.1 Discrete Choice Travel Cost Model

The two prices for an outdoor recreation trip defined in Section 3 can be used in a travel cost model to find the WTP of a local recreation trip (when using the price defined in Equation 7) or the accounting value of a trip (price from Equation 6). I estimate my model for multiple time periods, allowing me to observe trends in both values through time. I follow McFadden's binary choice random utility model (McFadden, 1974a,b). I model individuals making a discrete choice to take an outdoor recreation trip or choose a "no deviation" option where they participate in all daily activities other than the outdoor recreation trip. The utility for an individual $i$ choosing to take her utility-maximizing number of trips (denoted as $j$ and equal to either one or zero) is

$$U_i^j = V_i^j + \varepsilon_i$$

where $V$ is the deterministic utility that is a function of observables and $\varepsilon$ is the error term that I assume follows a Type 1 Extreme Value distribution. This distribution assumption allows me to use logistic choice probabilities to model choice-taking behavior.

I condition on three-year time periods and estimate the value of a local outdoor recreation trip for seven periods between 2003 and 2023, enabling the observation of trends in the value provided by outdoor recreation through time.

I model the conditional indirect utility of taking a trip ($j = 1$) in time period $t$ as

$$V_i^1 \mid t = \beta_0 + \beta_1 \pi_i + \beta_2 D_i + \beta_3 y_i + \beta_4 \mathbf{M_i} + \beta_5 \mathbf{L_i} \tag{8}$$

where $\beta_0$ is the baseline utility of taking a trip, $\pi$ is the travel cost of a trip ($\pi^{\text{WTP}}$ or $\pi^{\text{acc.}}$), $D$ is the quality of the trip measured as the activity participated in, $y$ is income measured in 2023 USD, $\mathbf{M}$ is a vector of demographic characteristics, and $\mathbf{L}$ is a vector of trip costs for substitute leisure activities that are not outdoor recreation. I normalize the conditional indirect utility of not taking a trip to zero, $V^0 = 0$. Choosing to normalize to zero over any arbitrary constant has no effect on estimating the value for a local recreation trip other than easing the estimation of parameters, as it is eventually differenced out.

I convert travel time to travel cost, $\pi$, in the two previously discussed ways: using one-third the wage rate or a replacement wage. Therefore, I estimate Equation 8 separately for each method. For both, I assume the marginal expenditure on market goods for local trips is minimal, and set $p_i x_i = 0$. To calculate the travel cost in a welfare setting where the value per trip is the WTP per trip, I use one-third of an individual's wage rate, such that Equation 7 can be rewritten as

$$\pi_i^{\text{WTP}} = \frac{1}{3} w \, t_i^{\text{near}}.$$

Second, when calculating the accounting value of a trip, I convert travel time to a monetary cost using a replacement wage. Theoretically, the conceptual framework suggests pricing time at the wage one could earn on the market for a similar activity, such as the \$15 per hour median wage for taxi drivers (O*NET Online, 2024), because travel time in this setting is time spent driving. However, to maintain consistency with national accounts, I instead use the Bureau of Labor Statistics' national median hourly wage of \$16.08 for maids and housekeeping cleaners (SOC 37-2012), the occupation category used by the BEA to value all activities in the household production satellite accounts (Bureau of Labor Statistics, 2024; Bureau of

Economic Analysis, 2018a). Using this single replacement wage allows my estimates to be directly comparable to BEA's valuations of household production. Therefore, Equation 6 becomes

$$\pi_i^{\text{acc.}} = 16.08 \times t_i^{\text{near}}.$$

The choice probability of taking a trip is modeled as

$$\Pr(j = 1) = \frac{\exp(V^1)}{\exp(V^0) + \exp(V^1)} = \frac{\exp(V^1)}{1 + \exp(V^1)}$$

and parameters are estimated by maximizing the log of the likelihood of taking a trip versus not deviating.

I calculate an individual's value for a daily trip to a local recreation site by finding the difference between the expected utility of taking a trip $\mathbb{E}[U^1]$ and the expected utility of never taking a trip $\mathbb{E}[U^0]$, where the expected utilities can be calculated using the logged sum of indirect utilities. The change in expected utility is converted to monetary units by using the marginal utility of money parameter, $\beta_1$. The average value of a single trip in time period $t$ is

$$v_t = -\frac{1}{\beta_1}\left(\mathbb{E}[U^1] - \mathbb{E}[U^0]\right)$$

which can be rewritten using the log-sum formulas as

$$v_t = -\frac{1}{\beta_1}\left(\ln \sum_j e^{V_i^j} - \ln e^{V^0}\right) = -\frac{1}{\beta_1} \ln \sum_j e^{V_i^j}$$

and expanding the summation term yields

$$v_t = -\frac{1}{\beta_1} \ln \left(e^{V_i^1} + e^{V^0}\right) = -\frac{1}{\beta_1} \ln \left(e^{V_i^1} + 1\right). \tag{9}$$

After estimating the parameters in Equation 8, I use predicted indirect utility levels $V_i$ and the $\beta_1$ to calculate the average value of a trip for all individuals using both travel cost prices, $\pi^{\text{WTP}}$ and $\pi^{\text{acc.}}$.

## 5.2 Aggregation

The ATUS sampling strategy is designed to produce national, annual statistics. The sampling strategy is not designed to be subset to geographic units below the national level or time periods shorter than one year, although it can be subset to demographic groups of interest. I use the sample weights to calculate the national value for annual access to local recreation. To do so, I multiply each individual's value of a trip, $v_t$, by their ATUS sampling weight. The national value for annual access to local recreation is

$$\Upsilon_t^{\text{national}} = \sum_i \omega_i v_t \tag{10}$$

where $\omega$ is the sampling weight. I bootstrap the standard errors for $\Upsilon_t$. The time period $t$ is the period the data is conditioned on to parameterize Equation 8.

In addition to finding the average value per trip ($v_t$) and the national value per year ($\Upsilon_t^{\text{national}}$), I calculate the annual value of access to local outdoor recreation per person ($\Upsilon_t^{\text{personal}}$). To do so, I follow the American Time Use Survey's methodology for calculating the expected number of trips to be taken by an individual annually, $\mathbb{E}[x]$ (U.S. Department of Labor and Bureau of Labor Statistics, 2022). I find the average annual

number of trips for three-year time periods. I multiply the expected number of trips by the expected value per trip for the relevant three-year period,

$$\Upsilon_t^{\text{personal}} = \mathbb{E}[x_i \mid t]\,\mathbb{E}[v_t \mid t] \tag{11}$$

Calculating the annual value per capita and nationally using this aggregation method assumes that local outdoor leisure is a homogeneous good after controlling for quality and the availability of substitutes (Addicott and Fenichel, 2019). If my measure of quality at sites and travel time to substitute leisure activities properly control for quality and availability of substitutes, my estimate of the expected national WTP will equal the aggregated changes in value for heterogeneous local outdoor leisure trips.

I estimate $\Upsilon_t^{\text{national}}$, $\Upsilon_t^{\text{personal}}$ and $v_t$ using a three-year period $t$ to track changes in welfare through time. The $t$ subscript is the middle year of the three-year period. This allows for the supply of outdoor leisure opportunities and alternative leisure substitutes to change over time, despite not directly observing the land and water that enable outdoor recreation. Demand will quickly adapt to supply changes, so conditioning on a three-year period allows me to estimate the number of trips demanded based on that period's recreation opportunities and alternatives. Conditioning on a single year results in sample sizes too small to gain informative statistical insight. I condition on income quintiles and race to test how my estimates change across populations of interest.

## 5.3 Matching and Identification

I use the heterogeneity in individuals' locations and the local outdoor sites they visit to identify my parameters. This provides more variation in the travel time, and thus travel cost, than is typically observed when estimating demand (Burt and Brewer, 1971). I directly observe the travel time for individuals who take a trip to a local recreation site (recreators).

Using nearest-neighbor matching, I construct expected travel time for individuals who do not take a trip (non-recreators). For each non-recreator, I match them to $n$ recreators where $n \in \{2, 3, 5\}$. Non-recreators are only matched with recreators that are in their same geographic state and time period. To find matches, I calculate propensity scores using the following three equations

$$P_i = \alpha_1 H_i^S + \alpha_3 H_i^W + \alpha_3 H_i^E + \varepsilon_i \tag{12}$$

$$P_i = \alpha_1 H_i^S + \alpha_3 H_i^W + \alpha_3 H_i^E + \alpha_4 B_i + \varepsilon_i \tag{13}$$

$$P_i = \alpha_1 H_i^S + \alpha_3 H_i^W + \alpha_3 H_i^E + \alpha_4 B_i + \alpha_5 R_i + \varepsilon_i \tag{14}$$

where $H^S$ is the time an individual spent sleeping the day they were interviewed for the ATUS, $H^W$ is the time spent working, $H^E$ is time spent eating or socializing. These $H$ variables measure how an individual spends their time throughout the day. $B$ is an indicator variable for if an individual is interviewed on a weekend or holiday and $R$ is an indicator variable for if an individual lives in a rural area. Using Euclidean distance, I match non-recreators to their $n$ nearest neighbors using the difference between the non-recreators' propensity score $P_i$ and the recreators'. This is done using the `MatchIt` package in R (Greifer, 2025). I use the average travel time of a non-recreator's nearest $n$ neighbors to impute travel time for non-recreators.

In addition to varying the number of matches $n$, I impose caliper restrictions of 0.05, 0.15, or 0.25 Euclidean distance between the recreator's and non-recreator's propensity scores. A caliper restricts matches to recreators whose propensity scores fall within the specified distance of the non-recreator's score. This

means a non-recreator is matched to up to $n$ recreators, but only those whose propensity scores lie within the caliper threshold. If fewer than $n$ recreators meet this criterion, the non-recreator is matched to fewer neighbors; if no recreators fall within the caliper, the non-recreator is excluded from the analysis. This approach prioritizes match quality over quantity, ensuring that imputed travel times are based on sufficiently similar individuals.

I measure recreators' quality of local outdoor leisure trips with the outdoor activity they most likely participate in, assuming the quality of trips closely follows an individual's activity. For recreators, I directly observe the activity. For non-recreators, I impute their expected activity using their $n$ nearest neighbors' activities. I use this expected activity as the approximation for the expected quality of a trip to non-recreators.

I use the lower bound of income bins reported in the ATUS to measure all individuals' income. Income is binned, so $\hat{\beta}_y$ is set identified (Manski and Tamer, 2002). I use this same income when I use one-third of an individual's wage rates to price the opportunity cost of travel time.

I observe the reported racial group of all individuals in the ATUS. I group reported race and ethnicity into the following categories: non-Hispanic White, Black, Hispanic, Asian, or other.[1]

To control for the availability of substitute alternative leisure activities, I include a vector of the expected travel times for the six alternative leisure activities presented in Table 2. I calculate the expected travel time for an individual $i$ by finding the average travel time for each leisure activity in that individual's county that year.

## 5.4 Inferring Natural Capital Stock Changes

My estimates of the value of a recreation trip (both accounting and WTP) reflect that value at the time of each survey, capturing both income effects and changes in ecosystem service provision that occurred during the 2003–2023 period. Recent theoretical work on the valuation of ecosystem services suggests that WTP estimates should be adjusted over time to account for relative price changes (RPCs) that arise from the limited substitutability between environmental amenities and market goods (Drupp et al., 2025b). The income elasticity framework implies that as national income grows, the real value of environmental amenities should increase at a rate given by $\text{RPC}_t = \eta(g_C - g_E)$, where $\eta$ is the income elasticity of WTP, $g_C$ is the growth rate of consumption measured by GDP growth, and $g_E$ is the growth rate of ecosystem services.

I leverage this framework to conduct a back-of-the-envelope calculation that infers changes in the stock of recreation-enabling assets over the 20-year study period. Rearranging the RPC formula yields

$$g_E = g_C - (\text{RPC}_t/\eta). \tag{15}$$

I calculate $\text{RPC}_t$ as the proportional change in estimated values over time:

$$\text{RPC}_t = \frac{v_t - v_{t-1}}{v_{t-1}} \tag{16}$$

where $v_t$ is the average value per trip in period $t$. Because all values are expressed in real 2023 USD, this captures the real relative price change. I take $\eta$ from published estimates in Drupp et al. (2025b) and use observed U.S. GDP growth as $g_C$. With these three components observed or estimated, I solve for $g_E$:

[1] While individuals certainly identify with more specific racial or ethnic groups than these categories, there are not enough observations to subset the variable further without losing the ability to make useful statistical inferences.

Table 3: **Average Annual Value for Local Outdoor Recreation, 2003–2023.** This table reports the mean value across all time periods for the three welfare measures shown in Figure 2: value per trip ($v_t$), annual value per person ($\Upsilon_t^{\text{personal}}$), and national annual value ($\Upsilon_t^{\text{national}}$). Results are shown for both travel time pricing methods: one-third of the individual's wage rate and the $16.08 replacement wage. All values are in real 2023 USD.

| Travel Cost Method | Value per Trip (2023 USD) | Value per Person (2023 USD) | Value Nationally (Billion 2023 USD) |
|---|---|---|---|
| replacement_wage | 15.87 | 669.22 | 174.52 |
| third_wage | 18.47 | 780.86 | 203.45 |

the implied growth rate of the recreation-enabling asset stock. This approach allows me to infer whether the dispersed infrastructure that enables local outdoor recreation (*i.e.,* parks, greenways, tree-lined streets, mixed-use spaces) has been maintained, expanded, or depleted during 2003–2023, addressing a key challenge in natural capital accounting when observing changes in dispersed natural assets is difficult.

# 6 Results

Figure 2 shows the trend in the value for local recreation in three different ways: a) the average U.S. resident's value per local recreation trip ($v_t$) defined in Equation 9, b) the average annual value per capita ($\Upsilon_t^{\text{personal}}$), which accounts for changes in the average number of trips taken per year, defined in Equation 11 and c) the national value for local recreation ($\Upsilon_t^{\text{national}}$) which accounts for changes in the average number of trips and population growth, defined in Equation 10. Results are shown for my preferred model, which uses Equation 14 to calculate propensity scores, matches non-recreators to their five nearest neighbors, and uses a caliper of 0.05 for nearest-neighbor matches. Regression results for my preferred model are in Appendix Section A.3. The results are robust to different imputation methods, including the matching equation, the number of matches and the caliper restriction (Appendix Section A.4, Figures 8 and 9).

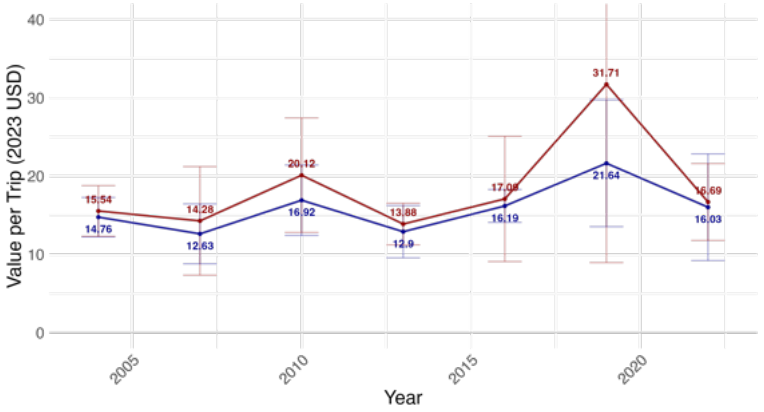
Figure 2 shows that when aggregating to the national level rather than to a sub-national demographic group, the use of a replacement wage versus one-third of the average wage rate does not have major implications. The average value per trip has remained around $16 real 2023 U.S. dollars (hereafter, $) if using a replacement wage and $18 if using one-third of the wage rate (Table 3). The two notable exceptions are the 2009-2011 and 2018-2020 periods. In each, the average value sharply increases. However, this increase also comes with significant variation, and neither increase significantly differs from the average value per trip over the entire study period. The change in value in these periods is more notable when using one-third of the wage rate than when using a replacement wage.

The average value over the entire study period for annual access to local recreation per person is $670 when using a replacement wage and $781 when using one-third of the wage rate (Table 3). The annual value per person has not grown or decreased over the past 20 years in a statistically significant way, regardless of pricing method, although there is an upward trend in point estimates likely due to recreators taking more trips per year (see Figure 1). Over the study period, the average national value for annual access to local parks is $175 billion when using a replacement wage and $203 billion when using one-third of the wage rate. The national value for annual access has increased by $96 ($102) billion from $136 ($129) billion in the first period of 2003 – 2005 to $241 ($231) in the last period of 2021 – 2023 when using one-third of the wage rate (a replacement wage). Both increases are significant at the $\alpha = 0.1$ level, but not at $\alpha = 0.05$.

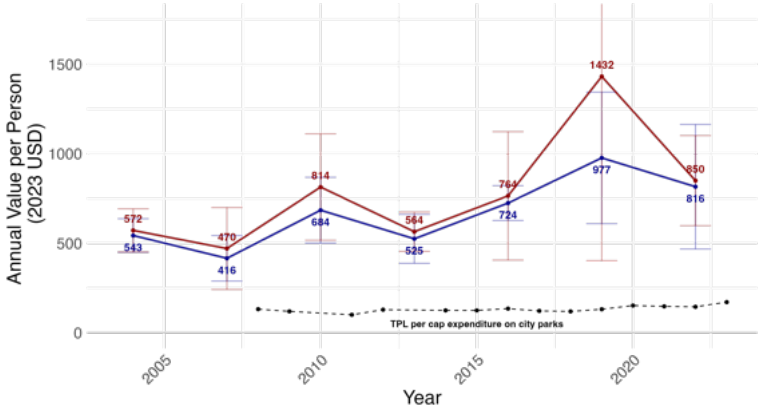
The use of a replacement wage versus one-third of the average wage rate has major implications when

Figure 2: **The 20-Year Trend in Annual Value for Local Outdoor Recreation, 2003–2023.** Panel (a) shows the average value per local recreation trip ($v_t$, Equation 9), Panel (b) shows the annual value of access to local recreation per average U.S. resident ($\Upsilon_t^{\text{personal}}$, Equation 11), and Panel (c) shows the annual national value for access to local recreation ($\Upsilon_t^{\text{national}}$, Equation 10), which accounts for population growth. Each panel compares two travel time pricing methods: one-third of the individual's wage rate (red) and the $16.08 replacement wage (blue). Error bars represent 95% confidence intervals. Estimates are conditioned on three-year periods to ensure adequate sample sizes; results are plotted at the midpoint year (e.g., 2003–2005 data plotted at 2004). All values are in real 2023 USD.

(a) Value of a Local Recreation Trip ($v_t$)




(b) Annual Value of Access to Local Recreation per Person ($\Upsilon_t^{\text{personal}}$)




(c) Annual Value for Access to Local Recreation, Nationally ($\Upsilon_t^{\text{national}}$)

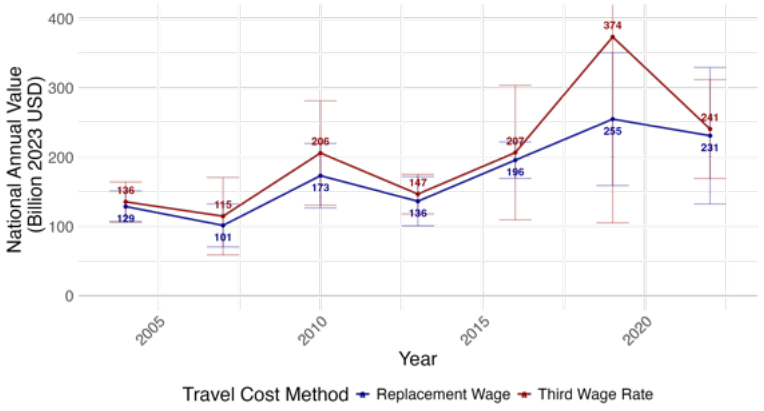
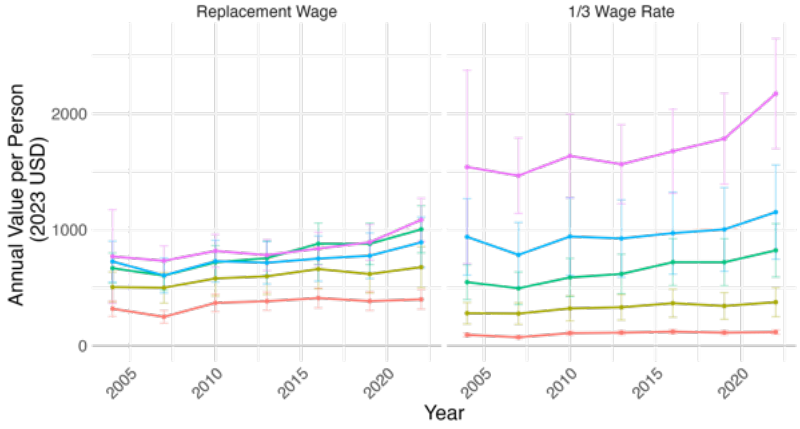
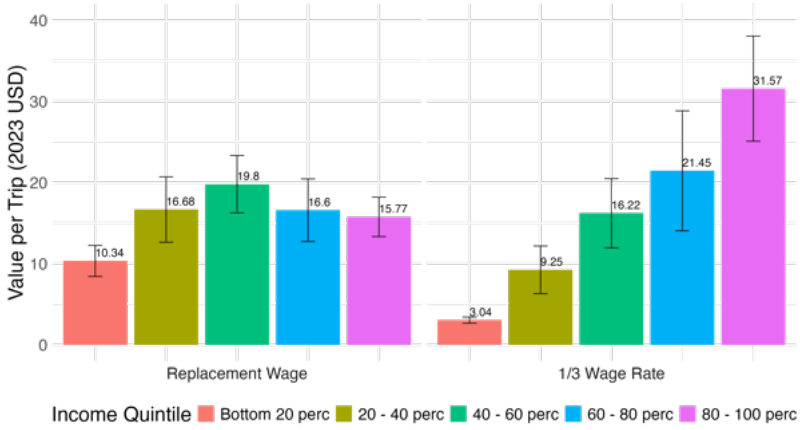


Figure 3: **Annual Value for Local Recreation by Income Quintile.** The figures show national trends in average annual value per person and the average value per trip (which is average across all time periods). This figure differs from above, in that it uses time-varying trips, but a constant value per trip to calculate national annual value for access to local recreation.

(a) Trend in National Value of Access to Local Recreation




(b) Avg Value per Trip




estimating the average value for a recreation trip across different economic classes (Figure 3). Regardless of the method, the richest quintile has a significantly higher value per trip than the poorest income quintile. However, using a replacement wage reduces the difference in the richest quintile's value from 10 times larger than the poorest to 1.5 times larger.

After controlling for income, there are significant differences in different racial and ethnic groups' value for a local recreation trip (Appendix Section A.2, Figure 7). Regardless of how I price travel time, Hispanic people's value per trip is double all other racial groups'. However, because of variation in the value per trip across Hispanic people, this higher value is not statistically different from any other racial group. All other racial groups' values are more precisely estimated and not significantly different from each other.

# 7 Back-of-the-Envelope: Changes in Recreation-Enabling Natural Capital

Beyond documenting trends in recreation values, I use the framework developed by Drupp et al. (2024, 2025b) to infer whether the stock of recreation-enabling assets has been maintained, expanded, or depleted over the 20-year study period. As described in Section 5.4, I calculate the implied growth rate of recreation-enabling assets ($g_E$) using the formula $g_E = g_C - (\text{RPC}_t/\eta)$, where $g_C$ is consumption growth approximated by U.S. GDP growth, $\text{RPC}_t$ is the relative price change in recreation values (Equation 16), and $\eta$ is the income elasticity of the willingness to pay for ecosystem services. I use three elasticity values from Drupp et al. (2025b) for sensitivity: $\eta = 0.6$ (general ecosystem services), $\eta = 0.66$ (forest ecosystem services), and $\eta = 1.0$ (unity).

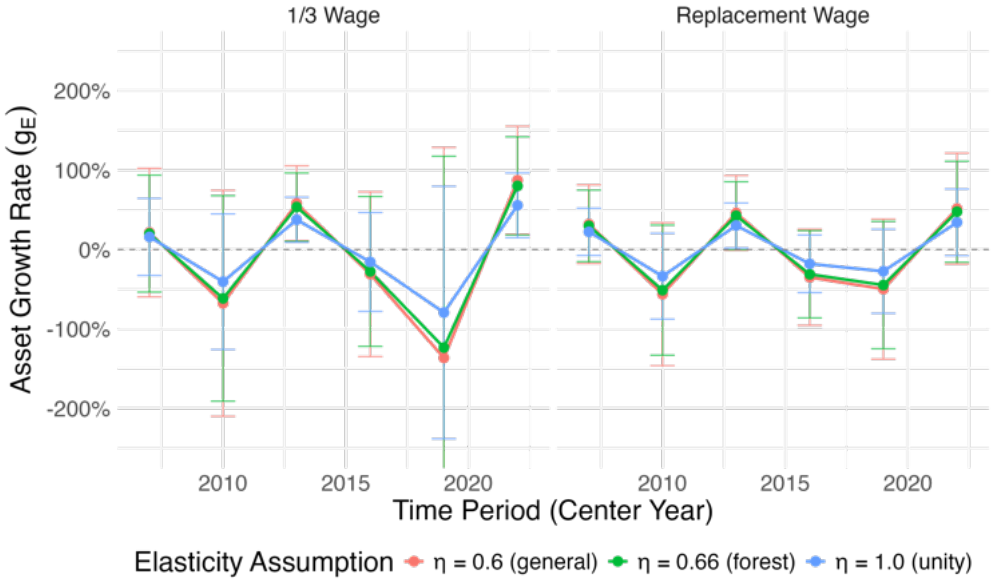
The enabling asset for outdoor recreation has not significantly increased nor decreased over my time period. Figure 4 shows the estimated growth rates of recreation-enabling assets ($g_E$) over time, separately for the two time pricing methods. For every period where $g_E$ has decreased there is a period where $g_E$ has similarly increased. On average over my study period, $g_E$ is close to zero (where close considers the large confidence intervals). When using one-third the wage rate to price travel time, $g_E$ is on average: -11 percent ($\eta = 0.6$), -9 percent ($\eta = 0.66$), -4 percent ($\eta = 1$). When using a replacement wage, $g_E$ is closer to zero: -1.8 percent ($\eta = 0.6$), -1.1 percent ($\eta = 0.66$), 1.4 percent ($\eta = 1$). The difference is likely due to the 2018–2020 period, where variation in income led to more volatility in the value per trip calculated using one-third the average wage rate (see Figure 2).

To interpret increases and decreases in $g_E$, it is helpful to review the components of Equation 15: $g_E = g_C - (\text{RPC}_t/\eta)$. First, when RPC = 0 (no change in the value per trip), the enabling asset grows at the same rate as consumption ($g_E = g_C$). Second, a negative RPC, meaning the value per trip declined, implies the enabling asset for recreation became relatively less scarce and $g_E > g_C$. Third, a positive RPC implies increased scarcity and $g_E < g_C$. Finally, the income elasticity $\eta$ amplifies these effects: smaller elasticities ($\eta < 1$) magnify the RPC term, producing more volatile $g_E$ estimates.

Consider the 2012–2014 period using the replacement wage method. GDP growth was $g_C = 6.6$ percent, and the value per trip fell 24 percent from \$17 to \$13 (i.e., RPC $= -0.24$). This negative RPC indicates recreation became less scarce, pushing $g_E$ well above $g_C$: between 30 percent ($\eta = 1$) and 46 percent ($\eta = 0.6$).

These results highlight a key insight from the natural capital accounting framework: rising willingness to pay for an ecosystem service can signal either increased environmental quality or increased scarcity. The value per trip trends in Figure 2 alone do not reveal whether the enabling asset for outdoor recreation grew or shrank. These trends must be combined with income growth to infer asset stock changes. When doing so, I find that the natural capital enabling local outdoor recreation has been roughly maintained over the past two decades. Despite substantial period-to-period volatility, periods of apparent expansion offset periods of contraction, resulting in near-zero average growth in $g_E$. This suggests that, in aggregate, the infrastructure and natural areas enabling local recreation have kept pace with growing national demand (see Figure 2, Panel C), a finding consistent with sustainable management of this recreational asset at the national level.

Figure 4: **Implied Growth Rate of Recreation-Enabling Assets.** The figure shows the estimated growth rate of the natural capital stock that enables local outdoor recreation ($g_E$) for each three-year period from 2007–2023. Estimates are calculated using the formula $g_E = g_C - (\text{RPC}_t/\eta)$ where $g_C$ is GDP growth (shown as black diamonds), $\text{RPC}_t$ is the relative price change in recreation values, and $\eta$ is the income elasticity. Three elasticity values are shown: 0.6 (general ecosystem services), 0.66 (forest services), and 1.0 (unity). Error bars represent 95% confidence intervals. Positive $g_E$ indicates asset expansion; negative $g_E$ indicates contraction. The two panels show results separately for one-third wage (left) and replacement wage (right) time pricing methods.




# 8 Discussion

This paper provides repeated, national-scale estimates of the value of local outdoor recreation in the United States over two decades using the ATUS. The average value per trip is stable at roughly \$16–\$18 (2023 USD), with upticks in value around the 2008 financial crisis and the 2020 COVID-19 pandemic. The increase in value per trip was temporary and not statistically significant, however the number of trips taken by the poorest quintile permanently increased after the 2008 financial crisis, and a similar increase is found for the richest quintile following the COVID-19 pandemic (Figure 1a). After scaling by trips and population, the national annual value of local recreation is \$231–\$241 billion in 2023, and has increased by roughly \$100 billion since the early 2000s.

This paper also describes who benefits. Participation and trip taking are widespread across income and racial groups. The national value trend is similar regardless of how time is priced. However, when analyzing who benefits (rather than the national aggregate), the modeling choice of how to price travel time has consequence. This modeling choice substantially changes the difference in the value per trip of the richest income quintile compared to the poorest from 10 times larger to only 1.5 times. This result highlights that the modeler's choice of pricing travel time has major implications for interpreting who prefers outdoor recreation. Using the traditional methodology (one-third of the average wage rate) may lead to concluding that recreation trips are a luxury good, where the demand rises quickly with income. However, when using the replacement wage approach common in household production accounting, the difference in the value of

a trip between the richest and poorest quintile is less than the difference in their income. After controlling for income, the choice of how to price travel time does not alter the qualitative ordering across racial groups conditional on income.

The results have implications for policy and environmental-economic statistics. First, because the ATUS supports annual repetition, the value of a local recreation trip can be easily and repeatedly estimated through time. The stability in value per trip but growth in national value (via trips and population) highlights the importance of tracking both intensive and extensive margins. Second, for distribution-sensitive benefit-cost analyses and related guidance, the choice of time-pricing matters. Using a replacement wage produces distributional summaries that depend less on labor-market earnings disparities and more on the realized service produced by an individual (in this case, the value of the travel service actually produced rather than the foregone wage income). This may be attractive when summarizing who benefits from public investments (e.g., neighborhood parks, trails) without embedding wage-based income disparities into benefit estimation (Ando et al., 2024). Finally, pricing travel time at a replacement wage aligns recreation travel squarely within household production accounts. Incorporating the methods from this paper into national satellite accounts could complement the Outdoor Recreation Satellite Account's market reclassifications and expand the Household Production Satellite Account. In turn, this provides a bridge from recreation demand to non-market accounting practices, advancing the integration of environmental benefits from recreation into new national environmental-economic statistics.

Finally, this paper provides a method for inferring changes in the natural capital stock that enables outdoor recreation, an asset that is difficult to observe directly. The "enabling asset" for local recreation is a composite of parks, trails, green spaces, water bodies, and the infrastructure that provides access to them. Measuring changes in this stock directly would require comprehensive spatial inventories of recreation opportunities and their quality over time, data that do not exist at the national scale. By combining repeated estimates of the value per trip with consumption growth rates, I use the framework from Drupp et al. (2024) to back out the implied growth rate of this enabling asset ($g_E$). My estimates suggest that, on average, the enabling asset for local outdoor recreation has neither grown nor declined over the past two decades. This finding is consistent with sustainable management of recreation-enabling natural capital at the national level: supply has roughly kept pace with demand. This approach offers a practical tool for natural capital accounting and sustainable development measurement, providing a way to track whether the assets underlying non-market ecosystem services are being maintained, expanded, or depleted, even when the assets themselves cannot be directly inventoried.

In sum, the paper demonstrates a scalable way to measure non-market recreation benefits over time, shows how distributional conclusions hinge on the valuation of time, offers an accounting-consistent path to embedding recreation benefits in national statistics and policy appraisal, and provides a method for inferring whether recreation-enabling natural capital is being sustainably managed.

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# A Appendix

## A.1 Detailed Activity Frequencies

Table 4: Total time spent over study period on outdoor activities.

| | Activity | Avg. Annual Hours (Million) | Activity Description | Actvity Category |
|---|---|---|---|---|
| 1 | 130131.00 | 2748.65 | Walking | Walking |
| 2 | 20603.00 | 1924.90 | Walking, exercising, playing with animals (2008+) | Pet Care and Leisure |
| 3 | 120101.00 | 814.43 | Socializing and communicating with others | Socializing and Relaxing |
| 4 | 130124.00 | 690.85 | Running | Running |
| 5 | 130112.00 | 656.55 | Fishing | Fishing |
| 6 | 130118.00 | 496.69 | Hunting | Hunting |
| 7 | 130104.00 | 429.85 | Biking | Biking |
| 8 | 120301.00 | 373.45 | Relaxing, thinking | Socializing and Relaxing |
| 9 | 130114.00 | 372.63 | Golfing | Golfing |
| 10 | 20601.00 | 358.74 | Care for animals and pets (not veterinary care) | Pet Care and Leisure |
| 11 | 110101.00 | 338.97 | Eating and drinking | Socializing and Relaxing |
| 12 | 130132.00 | 319.84 | Participating in water sports | Water sports |
| 13 | 30103.00 | 283.06 | Playing with hh children, not sports | Child Care and Leisure |
| 14 | 130116.00 | 264.75 | Hiking | Hiking |
| 15 | 30110.00 | 263.37 | Attending hh children's events | Child Care and Leisure |
| 16 | 120201.00 | 206.03 | Attending or hosting parties/receptions/ceremonies | Socializing and Relaxing |
| 17 | 30109.00 | 193.75 | Looking after hh children (as a primary activity) | Child Care and Leisure |
| 18 | 130106.00 | 153.40 | Boating | Water sports |
| 19 | 130103.00 | 153.39 | Playing basketball | Basketball |
| 20 | 130199.00 | 143.36 | Playing sports n.e.c. | Sports |
| 21 | 130202.00 | 117.58 | Watching baseball | Baseball |
| 22 | 130126.00 | 110.09 | Playing soccer | Soccer |
| 23 | 20602.00 | 96.76 | Care for animals and pets (not veterinary care) (2008+) | Pet Care and Leisure |
| 24 | 130125.00 | 90.99 | Skiing, ice skating, snowboarding | Snow Sports |
| 25 | 130120.00 | 87.22 | Playing racquet sports | Racquet Sports |
| 26 | 30105.00 | 82.99 | Playing sports with hh children | Child Care and Leisure |
| 27 | 130213.00 | 81.42 | Watching football | Football |
| 28 | 30111.00 | 69.37 | Waiting for/with hh children | Child Care and Leisure |
| 29 | 130224.00 | 67.41 | Watching soccer | Soccer |
| 30 | 130127.00 | 62.31 | Playing softball | Softball |
| 31 | 40110.00 | 61.95 | Attending nonhh children's events | Child Care and Leisure |
| 32 | 140101.00 | 57.90 | Attending religious services | Religious Activity |
| 33 | 120307.00 | 56.24 | Playing games | Sports |
| 34 | 130122.00 | 54.84 | Rollerblading | Rollerblading |
| 35 | 130102.00 | 54.47 | Playing baseball | Baseball |
| 36 | 30112.00 | 46.52 | Picking up/dropping off hh children | Child Care and Leisure |
| 37 | 130110.00 | 41.01 | Participating in equestrian sports | Equestrian Sports |
| 38 | 130130.00 | 35.61 | Playing volleyball | Volleyball |
| 39 | 130113.00 | 34.78 | Playing football | Football |
| 40 | 130226.00 | 33.30 | Watching vehicle touring/racing | Vehicle Touring/Racing |
| 41 | 130299.00 | 33.26 | Attending sporting events, n.e.c. | Sports |
| 42 | 40109.00 | 32.40 | Looking after nonhh children (as primary activity) | Child Care and Leisure |
| 43 | 130225.00 | 32.13 | Watching softball | Softball |
| 44 | 120202.00 | 32.10 | Attending meetings for personal interest (not volunteering) | Socializing and Relaxing |
| 45 | 140102.00 | 27.15 | Participation in religious practices | Religious Activity |
| 46 | 130134.00 | 26.94 | Working out, unspecified | Sports |
| 47 | 40503.00 | 22.78 | Animal and pet care assistance for nonhh adults | Pet Care and Leisure |
| 48 | 130108.00 | 20.71 | Climbing, spelunking, caving | Climbing |
| 49 | 120399.00 | 19.63 | Relaxing and leisure, n.e.c. | Socializing and Relaxing |
| 50 | 40599.00 | 18.31 | Helping nonhh adults, n.e.c. | Adult Care and Leisure |
| 51 | 30504.00 | 15.45 | Waiting associated with helping hh adults | Adult Care and Leisure |
| 52 | 130203.00 | 13.19 | Watching basketball | Basketball |
| 53 | 40105.00 | 13.11 | Playing sports with nonhh children | Child Care and Leisure |
| 54 | 130227.00 | 11.44 | Watching volleyball | Volleyball |
| 55 | 130229.00 | 10.24 | Watching water sports | Water sports |
| 56 | 30101.00 | 10.23 | Physical care for hh children | Child Care and Leisure |
| 57 | 30501.00 | 8.34 | Helping hh adults | Adult Care and Leisure |
| 58 | 130222.00 | 8.33 | Watching running | Running |
| 59 | 130212.00 | 8.22 | Watching fishing | Fishing |
| 60 | 40199.00 | 7.81 | Caring for and helping nonhh children, n.e.c. | Child Care and Leisure |
| 61 | 40111.00 | 7.80 | Waiting for/with nonhh children | Child Care and Leisure |
| 62 | 130123.00 | 7.50 | Playing rugby | Rugby |
| 63 | 40112.00 | 7.22 | Dropping off/picking up nonhh children | Child Care and Leisure |
| 64 | 130210.00 | 6.66 | Watching equestrian sports | Equestrian Sports |
| 65 | 130219.00 | 6.15 | Watching rodeo competitions | Rodeo Competitions |
| 66 | 130218.00 | 6.13 | Watching racquet sports | Racquet Sports |
| 67 | 130204.00 | 5.60 | Watching biking | Biking |
| 68 | 40101.00 | 5.31 | Physical care for nonhh children | Child Care and Leisure |
| 69 | 140105.00 | 5.19 | Religious education activities (2007+) | Religious Activity |
| 70 | 130206.00 | 4.67 | Watching boating | Water sports |
| 71 | 120299.00 | 4.26 | Attending/hosting social events, n.e.c. | Socializing and Relaxing |
| 72 | 130117.00 | 4.01 | Playing hockey | Hockey |
| 73 | 60201.00 | 3.80 | Extracurricular club activities | Extracurricular club activities |
| 74 | 130214.00 | 3.79 | Watching golfing | Golfing |
| 75 | 30199.00 | 3.74 | Caring for and helping hh children, n.e.c. | Child Care and Leisure |
| 76 | 20699.00 | 3.63 | Pet and animal care, n.e.c. | Pet Care and Leisure |
| 77 | 129999.00 | 3.23 | Socializing, relaxing, and leisure, n.e.c. | Socializing and Relaxing |
| 78 | 50201.00 | 2.98 | Socializing, relaxing, and leisure as part of job | Socializing and Relaxing |
| 79 | 130216.00 | 2.07 | Watching hockey | Hockey |
| 80 | 50203.00 | 1.96 | Sports and exercise as part of job | Sports |
| 81 | 30599.00 | 1.80 | Helping household adults, n.e.c. | Adult Care and Leisure |
| 82 | 30402.00 | 1.68 | Looking after hh adult (as a primary activity) | Adult Care and Leisure |
| 83 | 139999.00 | 1.60 | Sports, exercise, and recreation, n.e.c. | Sports |
| 84 | 40106.00 | 1.36 | Talking with/listening to nonhh children | Child Care and Leisure |
| 85 | 130223.00 | 1.13 | Watching skiing, ice skating, snowboarding | Snow Sports |
| 86 | 40401.00 | 1.04 | Physical care for nonhh adults | Adult Care and Leisure |
| 87 | 60299.00 | 0.87 | Education-related extracurricular activities, n.e.c. | |
| 88 | 130221.00 | 0.84 | Watching rugby | Rugby |
| 89 | 40399.00 | 0.84 | Activities related to nonhh child's health, n.e.c. | Child Care and Leisure |
| 90 | 130121.00 | 0.69 | Participating in rodeo competitions | Rodeo Competitions |
| 91 | 49999.00 | 0.49 | Caring for and helping nonhh members, n.e.c. | Adult Care and Leisure |
| 92 | 30499.00 | 0.48 | Caring for household adults, n.e.c. | Adult Care and Leisure |
| 93 | 149999.00 | 0.46 | Religious and spiritual activities, n.e.c. | Religious Activity |
| 94 | 140103.00 | 0.40 | Waiting assoc w/religious and spiritual activities | Religious Activity |
| 95 | 30401.00 | 0.13 | Physical care for hh adults | Adult Care and Leisure |
| 96 | 40402.00 | 0.12 | Looking after nonhh adult (as a primary activity) | Adult Care and Leisure |
| 97 | 130220.00 | 0.06 | Watching rollerblading | Rollerblading |
| 98 | 39999.00 | 0.03 | Caring for and helping hh members, n.e.c. | Adult Care and Leisure |

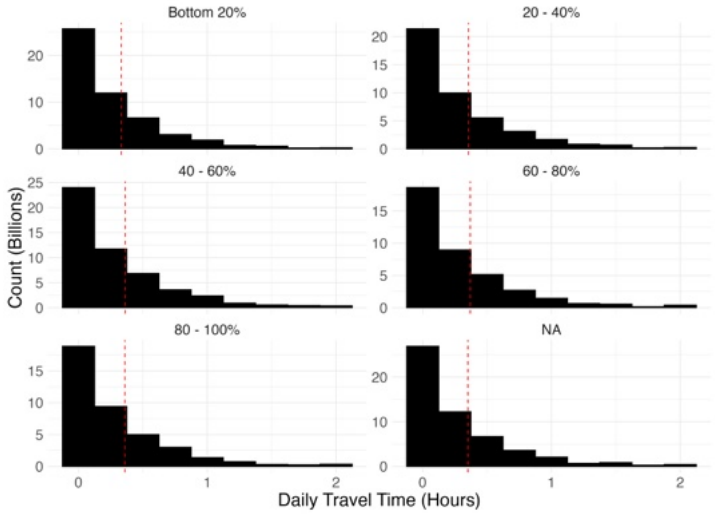
Table 5: Total time spent over study period on indoor activities, and their substitute category.

| | Activity Code | Avg. Annual Hours (Billion) | Activity Description | Substitute Category |
|---|---|---|---|---|
| 1 | 110101.00 | 442766.20 | Eating and drinking | Eating or Drinking |
| 2 | 70104.00 | 396287.31 | Shopping, except groceries, food and gas | Shopping |
| 3 | 120101.00 | 175489.42 | Socializing and communicating with others | Socializing |
| 4 | 120303.00 | 46299.01 | Television and movies (not religious) | Entertainment |
| 5 | 120403.00 | 45473.42 | Attending movies/film | Entertainment |
| 6 | 70103.00 | 38228.36 | Purchasing food (not groceries) | Eating or Drinking |
| 7 | 120201.00 | 36536.05 | Attending or hosting parties/receptions/ceremonies | Socializing |
| 8 | 130134.00 | 36534.70 | Working out, unspecified | Playing Sports or Games |
| 9 | 120499.00 | 31009.26 | Arts and entertainment, n.e.c. | Entertainment |
| 10 | 130133.00 | 28792.49 | Weightlifting/strength training | Playing Sports or Games |
| 11 | 120301.00 | 24903.76 | Relaxing, thinking | Hobby or Relaxing |
| 12 | 120307.00 | 23626.39 | Playing games | Playing Sports or Games |
| 13 | 120401.00 | 23546.02 | Attending performing arts | Entertainment |
| 14 | 120312.00 | 22598.20 | Reading for personal interest | Hobby or Relaxing |
| 15 | 130103.00 | 14489.81 | Playing basketball | Playing Sports or Games |
| 16 | 120202.00 | 12665.56 | Attending meetings for personal interest (not volunteering) | Hobby or Relaxing |
| 17 | 70101.00 | 11950.44 | Grocery shopping | Eating or Drinking |
| 18 | 130107.00 | 10631.03 | Bowling | Playing Sports or Games |
| 19 | 130199.00 | 10604.00 | Playing sports n.e.c. | Playing Sports or Games |
| 20 | 130128.00 | 10220.95 | Using cardiovascular equipment | Playing Sports or Games |
| 21 | 120402.00 | 9753.28 | Attending museums | Entertainment |
| 22 | 130203.00 | 8149.90 | Watching basketball | Entertainment |
| 23 | 130213.00 | 7950.46 | Watching football | Entertainment |
| 24 | 130202.00 | 7610.45 | Watching baseball | Entertainment |
| 25 | 130109.00 | 7539.08 | Dancing | Playing Sports or Games |
| 26 | 120308.00 | 7026.91 | Computer use for leisure (exc. Games) | Playing Sports or Games |
| 27 | 120306.00 | 6697.62 | Listening to/playing music (not radio) | Hobby or Relaxing |
| 28 | 130120.00 | 6518.63 | Playing racquet sports | Playing Sports or Games |
| 29 | 130105.00 | 6268.60 | Playing billiards | Playing Sports or Games |
| 30 | 110201.00 | 5504.41 | Waiting associated w/eating and drinking | Eating or Drinking |
| 31 | 130136.00 | 4351.67 | Doing yoga | Playing Sports or Games |
| 32 | 130101.00 | 3710.17 | Doing aerobics | Playing Sports or Games |
| 33 | 130130.00 | 3434.37 | Playing volleyball | Playing Sports or Games |
| 34 | 50202.00 | 3243.97 | Eating and drinking as part of job | Eating or Drinking |
| 35 | 120309.00 | 3211.55 | Arts and crafts as a hobby | Hobby or Relaxing |
| 36 | 130299.00 | 2790.40 | Attending sporting events, n.e.c. | Entertainment |
| 37 | 130224.00 | 2716.14 | Watching soccer | Entertainment |
| 38 | 130216.00 | 2365.76 | Watching hockey | Entertainment |
| 39 | 120305.00 | 2259.75 | Listening to the radio | Hobby or Relaxing |
| 40 | 130226.00 | 1988.11 | Watching vehicle touring/racing | Entertainment |
| 41 | 130122.00 | 1880.95 | Rollerblading | Playing Sports or Games |
| 42 | 130225.00 | 1729.57 | Watching softball | Entertainment |
| 43 | 50201.00 | 1646.16 | Socializing, relaxing, and leisure as part of job | Socializing |
| 44 | 130119.00 | 1604.68 | Participating in martial arts | Playing Sports or Games |
| 45 | 120504.00 | 1189.52 | Waiting associated with arts and entertainment | Entertainment |
| 46 | 130117.00 | 1168.58 | Playing hockey | Playing Sports or Games |
| 47 | 130227.00 | 1152.33 | Watching volleyball | Entertainment |
| 48 | 130218.00 | 933.83 | Watching racquet sports | Entertainment |
| 49 | 130232.00 | 887.29 | Watching wrestling | Entertainment |
| 50 | 130210.00 | 639.36 | Watching equestrian sports | Entertainment |
| 51 | 130207.00 | 611.63 | Watching bowling | Entertainment |
| 52 | 130229.00 | 504.81 | Watching water sports | Entertainment |
| 53 | 120311.00 | 494.96 | Hobbies, except arts and crafts and collecting | Hobby or Relaxing |
| 54 | 130214.00 | 489.34 | Watching golfing | Entertainment |
| 55 | 120399.00 | 488.02 | Relaxing and leisure, n.e.c. | Hobby or Relaxing |
| 56 | 130135.00 | 437.61 | Wrestling | Playing Sports or Games |
| 57 | 120299.00 | 431.61 | Attending/hosting social events, n.e.c. | Socializing |
| 58 | 130222.00 | 420.36 | Watching running | Entertainment |
| 59 | 50203.00 | 388.51 | Sports and exercise as part of job | Playing Sports or Games |
| 60 | 120501.00 | 374.69 | Waiting assoc. w/socializing and communicating | Socializing |
| 61 | 139999.00 | 283.11 | Sports, exercise, and recreation, n.e.c. | Playing Sports or Games |
| 62 | 130115.00 | 278.67 | Doing gymnastics | Playing Sports or Games |
| 63 | 70201.00 | 276.13 | Comparison shopping | Shopping |
| 64 | 130219.00 | 259.62 | Watching rodeo competitions | Entertainment |
| 65 | 130209.00 | 258.06 | Watching dancing | Entertainment |
| 66 | 120313.00 | 255.17 | Writing for personal interest | Hobby or Relaxing |
| 67 | 120502.00 | 229.56 | Waiting assoc. w/attending/hosting social events | Socializing |
| 68 | 130205.00 | 226.41 | Watching billiards | Entertainment |
| 69 | 120310.00 | 101.64 | Collecting as a hobby | Hobby or Relaxing |
| 70 | 129999.00 | 96.59 | Socializing, relaxing, and leisure, n.e.c. | Socializing |
| 71 | 130215.00 | 87.39 | Watching gymnastics | Entertainment |
| 72 | 120503.00 | 83.59 | Waiting associated with relaxing/leisure | Hobby or Relaxing |
| 73 | 130206.00 | 76.03 | Watching boating | Entertainment |
| 74 | 110299.00 | 70.17 | Waiting associated with eating and drinking, n.e.c. | Eating or Drinking |
| 75 | 130217.00 | 70.11 | Watching martial arts | Entertainment |
| 76 | 130220.00 | 63.27 | Watching rollerblading | Entertainment |
| 77 | 130211.00 | 50.65 | Watching fencing | Entertainment |
| 78 | 130223.00 | 44.97 | Watching skiing, ice skating, snowboarding | Entertainment |
| 79 | 130111.00 | 40.48 | Fencing | Playing Sports or Games |
| 80 | 70199.00 | 31.75 | Shopping, n.e.c. | Shopping |
| 81 | 130231.00 | 30.66 | Watching people working out, unspecified | Entertainment |
| 82 | 79999.00 | 29.37 | Consumer purchases, n.e.c. | Shopping |
| 83 | 130230.00 | 26.98 | Watching weightlifting/strength training | Entertainment |
| 84 | 130221.00 | 22.58 | Watching rugby | Entertainment |
| 85 | 70299.00 | 11.88 | Researching purchases, n.e.c. | Shopping |
| 86 | 120199.00 | 9.57 | Socializing and communicating, n.e.c. | Socializing |
| 87 | 130201.00 | 8.51 | Watching aerobics | Entertainment |
| 88 | 120599.00 | 7.88 | Waiting associated with socializing, n.e.c. | Socializing |
| 89 | 130204.00 | 7.33 | Watching biking | Entertainment |
| 90 | 119999.00 | 1.69 | Eating and drinking, n.e.c. | Eating or Drinking |

## A.2 Supplementary Figures

Figure 5: Distribution of average travel time for outdoor and indoor activities by income quintile and race. Outdoor activities have an average travel time below 30 minutes, while indoor activities have an average travel time just over 30 minutes.

(a) Outdoor Avg. Travel Time by Income




(b) Outdoor Avg. Travel Time by Race

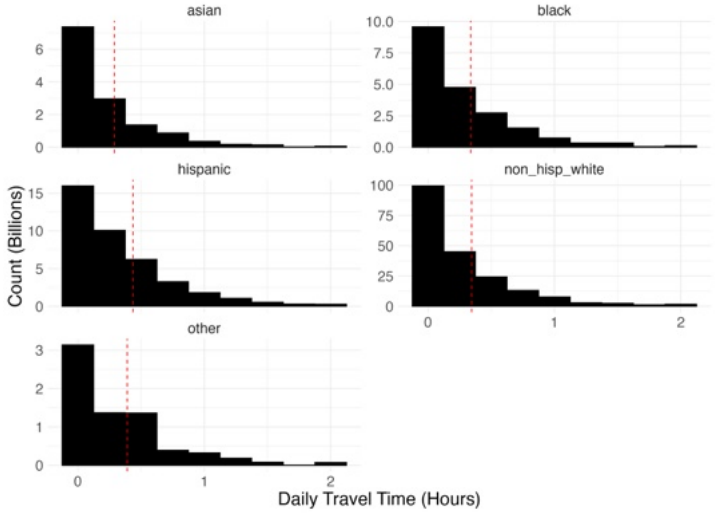
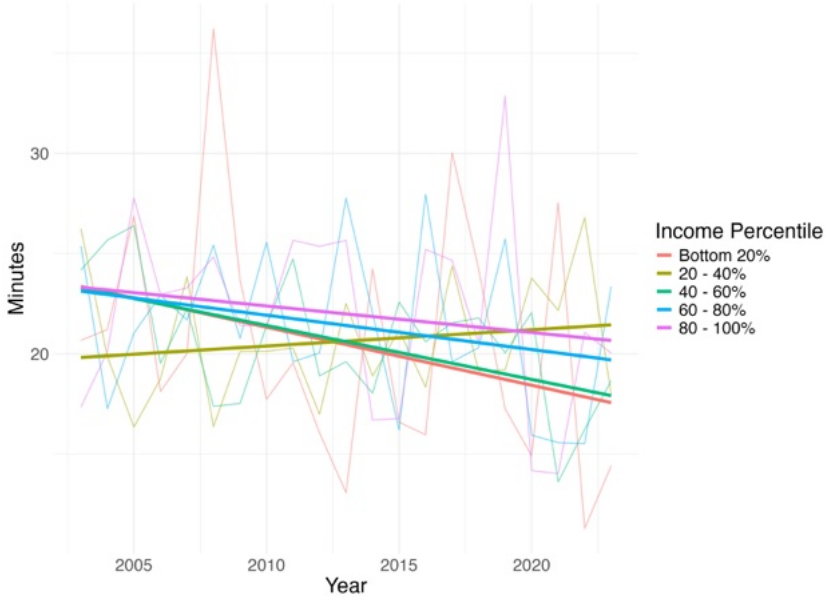


Figure 6: Average travel time for outdoor activities by income quintile and race.

(a) Outdoor Avg. Travel Time by Income




(b) Outdoor Avg. Travel Time by Race

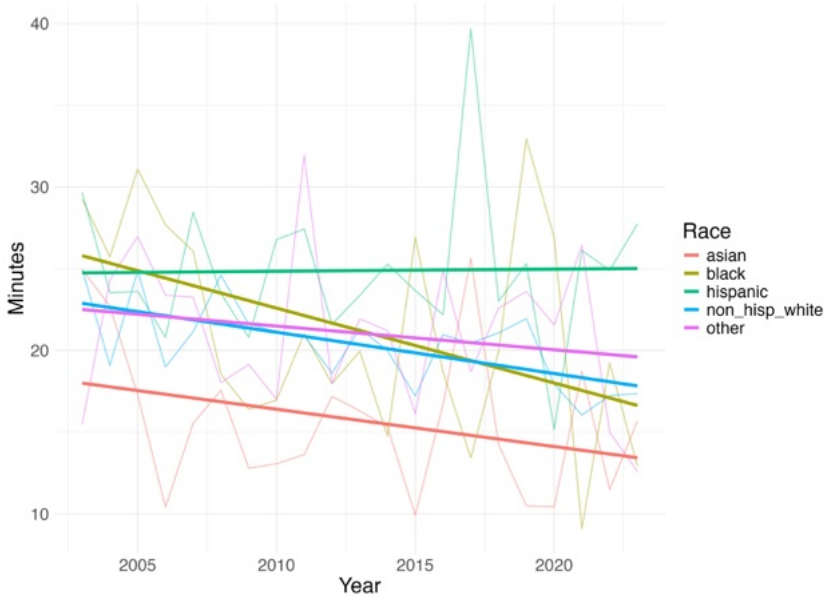
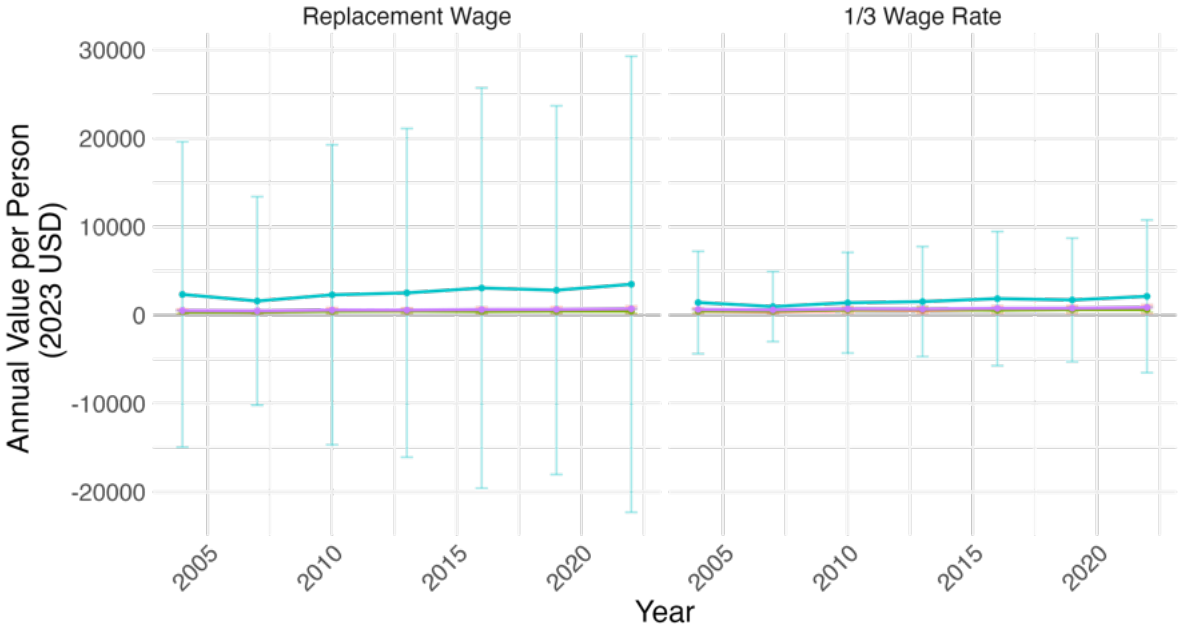
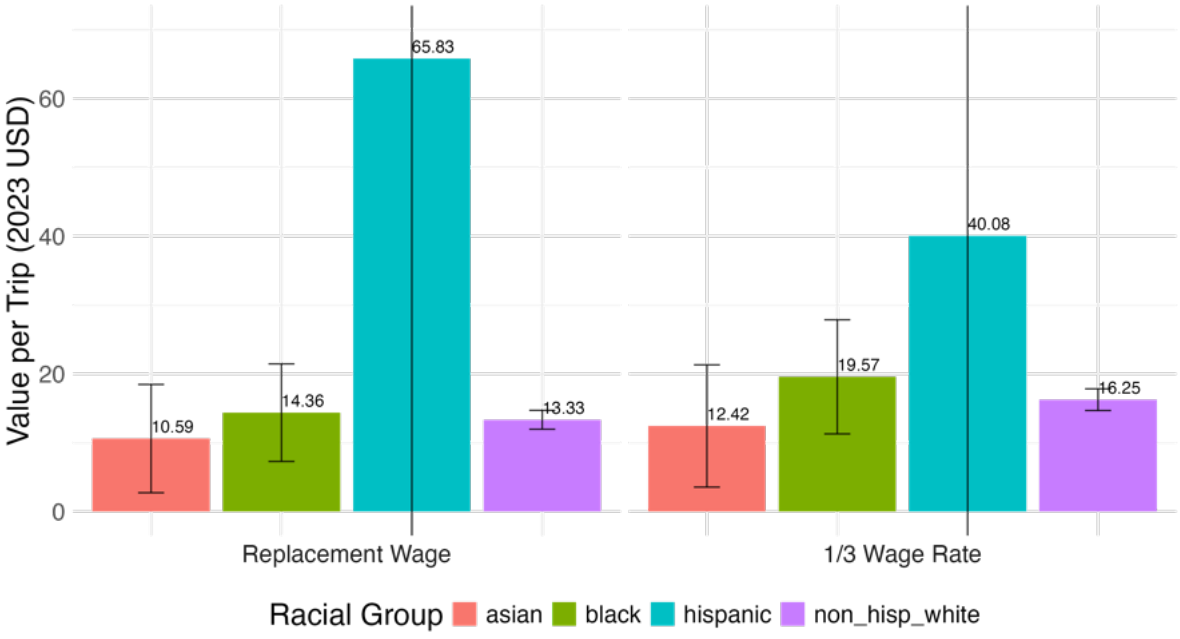


Figure 7: **Annual Value for Local Recreation by Race/Ethnicity.** After controlling for income, the use of a replacement wage or 1/3 wage rate does not significantly change the estimated value for local recreation for different racial groups. This contrasts with the results for income quintile, where the use of a replacement wage or 1/3 wage rate significantly changes the estimated value for local recreation.

(a) Trend in National value of Access to Local Recreation




(b) Avg Value per Trip



## A.3 Regression Results for Preferred Model

### A.3.1 One-third Wage Rate

Table 6: Three-year period regression results for my preferred model (5 neighbor matches, matched using Eqn. 14)

| | | 2003 | 2006 | 2009 | 2012 | 2015 | 2018 | 2021 |
|---|---|---|---|---|---|---|---|---|
| 1 | Dependent Var.: | trip | trip | trip | trip | trip | trip | trip |
| 2 | | | | | | | | |
| 3 | Constant | -1.147*** (0.1899) | -0.9261*** (0.1976) | -0.5510** (0.1876) | -0.4487* (0.1931) | -0.7005*** (0.1753) | -0.6829** (0.2205) | -0.6100** (0.2090) |
| 4 | travel_cost_third_mid | -0.0645*** (0.0050) | -0.0724*** (0.0069) | -0.0463*** (0.0063) | -0.0713*** (0.0071) | -0.0477*** (0.0069) | -0.0237** (0.0079) | -0.0728*** (0.0099) |
| 5 | prob_eat_drink | -1.823*** (0.0998) | -1.592*** (0.1223) | -1.368*** (0.1018) | -1.862*** (0.1213) | -1.668*** (0.1220) | -1.617*** (0.1332) | -1.980*** (0.1746) |
| 6 | prob_pet_care | -1.594*** (0.0543) | -1.522*** (0.0647) | -1.635*** (0.0560) | -1.626*** (0.0554) | -1.487*** (0.0553) | -1.438*** (0.0635) | -1.402*** (0.0671) |
| 7 | prob_child_care | -1.276*** (0.0656) | -1.233*** (0.0825) | -1.416*** (0.0716) | -1.707*** (0.0952) | -1.180*** (0.0841) | -1.005*** (0.1125) | -1.151*** (0.1311) |
| 8 | prob_social_relax_leisure | -1.481*** (0.0715) | -1.183*** (0.0946) | -1.846*** (0.0873) | -1.573*** (0.0968) | -1.882*** (0.1029) | -1.435*** (0.1098) | -1.557*** (0.1482) |
| 9 | prob_religous_spiritual | -1.051*** (0.2219) | -2.041*** (0.3602) | -1.716*** (0.2624) | -1.177*** (0.2594) | -1.114*** (0.2981) | -1.266** (0.4124) | -1.239** (0.4317) |
| 10 | prob_adult_care | -1.637*** (0.2605) | -0.8659* (0.4160) | -1.193*** (0.3584) | -1.613*** (0.2911) | -0.9021*** (0.2730) | -0.7874* (0.3069) | -0.7933* (0.3718) |
| 11 | fam_inc_mid_2023 | 6.59e-6*** (4e-7) | 7.25e-6*** (4.87e-7) | 5.12e-6*** (4.66e-7) | 5.68e-6*** (5.18e-7) | 4.3e-6*** (5.26e-7) | 3.1e-6*** (6.19e-7) | 6.59e-6*** (7.35e-7) |
| 12 | my_raceasian | -0.1784 (0.1118) | -0.2894* (0.1396) | -0.3772** (0.1176) | -0.0651 (0.1061) | -0.1597 (0.1124) | -0.1528 (0.1232) | -0.0750 (0.1255) |
| 13 | my_raceblack | -0.6216*** (0.0717) | -0.6219*** (0.0801) | -0.5168*** (0.0646) | -0.5035*** (0.0681) | -0.5325*** (0.0685) | -0.5973*** (0.0858) | -0.6078*** (0.1009) |
| 14 | my_racehispanic | -0.0434 (0.0584) | -0.3177*** (0.0700) | -0.1492** (0.0575) | -0.1378* (0.0633) | 0.1691** (0.0571) | -0.0992 (0.0713) | -0.0259 (0.0765) |
| 15 | my_raceother | 0.1067 (0.1153) | 0.3714** (0.1183) | -0.2368. (0.1344) | -0.0903 (0.1356) | -0.1561 (0.1475) | 0.1408 (0.1512) | -0.0005 (0.1620) |
| 16 | category_entertainment | 0.0071*** (0.0019) | -0.0009 (0.0022) | -0.0007 (0.0020) | -0.0040. (0.0022) | 0.0041. (0.0022) | -0.0063** (0.0024) | 0.0057** (0.0021) |
| 17 | category_shopping | -0.0029 (0.0059) | 0.0043 (0.0061) | 0.0066 (0.0054) | -0.0125* (0.0052) | -0.0089* (0.0045) | 0.0008 (0.0056) | 0.0004 (0.0059) |
| 18 | category_socializing | 0.0008 (0.0028) | -0.0029 (0.0031) | -0.0027 (0.0029) | 0.0073* (0.0032) | 0.0032 (0.0028) | 0.0012 (0.0031) | 0.0017 (0.0027) |
| 19 | category_eating_or_drinking | 0.0047 (0.0063) | -0.0054 (0.0068) | -0.0043 (0.0058) | 0.0062 (0.0060) | 0.0082 (0.0057) | 0.0042 (0.0061) | -0.0114. (0.0063) |
| 20 | category_playing_sports_or_games | 0.0002 (0.0022) | 0.0029 (0.0025) | -0.0033 (0.0024) | -0.0055. (0.0029) | 0.0008 (0.0024) | -0.0011 (0.0032) | -0.0015 (0.0035) |
| 21 | category_hobby_or_relaxing | 0.0016 (0.0018) | -0.0030 (0.0021) | 0.0034 (0.0021) | 0.0051** (0.0019) | -0.0004 (0.0021) | 0.0026 (0.0021) | 0.0025 (0.0020) |
| 22 | ---------- | ---------- | ---------- | ---------- | ---------- | ---------- | ---------- | ---------- |
| 23 | S.E. type | IID | IID | IID | IID | IID | IID | IID |
| 24 | Observations | 29,615 | 20,143 | 22,851 | 20,442 | 18,980 | 14,036 | 10,930 |
| 25 | Squared Cor. | 0.09573 | 0.08357 | 0.09390 | 0.10655 | 0.09277 | 0.07453 | 0.09678 |
| 26 | Pseudo R2 | 0.08956 | 0.08795 | 0.09302 | 0.10316 | 0.08842 | 0.07360 | 0.08593 |
| 27 | BIC | 22,416.4 | 14,698.9 | 17,994.2 | 15,724.2 | 15,650.4 | 11,686.6 | 9,440.6 |

### A.3.2 Replacement Wage

Table 7: Three-year period regression results for my preferred model (5 neighbor matches, matched using Eqn. 14)

| | | 2003 | 2006 | 2009 | 2012 | 2015 | 2018 | 2021 |
|---|---|---|---|---|---|---|---|---|
| 1 | Dependent Var.: | trip | trip | trip | trip | trip | trip | trip |
| 2 | | | | | | | | |
| 3 | Constant | -0.8518*** (0.1905) | -0.6430** (0.1984) | -0.3471. (0.1889) | -0.1743 (0.1935) | -0.5129** (0.1759) | -0.5632* (0.2211) | -0.3800. (0.2106) |
| 4 | travel_cost_15 | -0.0688*** (0.0050) | -0.0829*** (0.0070) | -0.0538*** (0.0059) | -0.0738*** (0.0063) | -0.0478*** (0.0060) | -0.0388*** (0.0070) | -0.0699*** (0.0082) |
| 5 | prob_eat_drink | -1.794*** (0.0998) | -1.516*** (0.1224) | -1.327*** (0.1020) | -1.788*** (0.1217) | -1.619*** (0.1226) | -1.553*** (0.1334) | -1.901*** (0.1751) |
| 6 | prob_pet_care | -1.598*** (0.0542) | -1.533*** (0.0646) | -1.640*** (0.0560) | -1.625*** (0.0554) | -1.489*** (0.0553) | -1.442*** (0.0635) | -1.410*** (0.0672) |
| 7 | prob_child_care | -1.257*** (0.0656) | -1.211*** (0.0825) | -1.391*** (0.0718) | -1.675*** (0.0953) | -1.149*** (0.0844) | -0.9480*** (0.1129) | -1.088*** (0.1317) |
| 8 | prob_social_relax_leisure | -1.460*** (0.0716) | -1.138*** (0.0948) | -1.796*** (0.0876) | -1.505*** (0.0971) | -1.826*** (0.1038) | -1.380*** (0.1100) | -1.486*** (0.1489) |
| 9 | prob_religous_spiritual | -0.9689*** (0.2217) | -1.941*** (0.3598) | -1.589*** (0.2630) | -1.047*** (0.2594) | -1.061*** (0.2988) | -1.234** (0.4128) | -1.130** (0.4319) |
| 10 | prob_adult_care | -1.626*** (0.2605) | -0.8450* (0.4139) | -1.150** (0.3592) | -1.597*** (0.2908) | -0.8673** (0.2729) | -0.7464* (0.3071) | -0.7412* (0.3725) |
| 11 | fam_inc_mid_2023 | 3.71e-6*** (3.43e-7) | 4.29e-6*** (4.08e-7) | 3.17e-6*** (3.85e-7) | 2.79e-6*** (4.34e-7) | 2.31e-6*** (4.41e-7) | 2.19e-6*** (5.38e-7) | 4.07e-6*** (6.61e-7) |
| 12 | my_raceasian | -0.1734 (0.1117) | -0.2918* (0.1395) | -0.3750** (0.1176) | -0.0573 (0.1060) | -0.1539 (0.1123) | -0.1479 (0.1233) | -0.0853 (0.1257) |
| 13 | my_raceblack | -0.6155*** (0.0718) | -0.6256*** (0.0803) | -0.5266*** (0.0648) | -0.5106*** (0.0684) | -0.5344*** (0.0686) | -0.5998*** (0.0859) | -0.6156*** (0.1011) |
| 14 | my_racehispanic | -0.0465 (0.0585) | -0.3225*** (0.0701) | -0.1504** (0.0576) | -0.1178. (0.0634) | 0.1766** (0.0571) | -0.0994 (0.0714) | -0.0246 (0.0766) |
| 15 | my_raceother | 0.1035 (0.1155) | 0.3738** (0.1186) | -0.2377. (0.1343) | -0.0783 (0.1358) | -0.1550 (0.1476) | 0.1442 (0.1515) | -0.0052 (0.1622) |
| 16 | category_entertainment | 0.0071*** (0.0019) | -0.0009 (0.0022) | -0.0008 (0.0020) | -0.0040. (0.0022) | 0.0042. (0.0022) | -0.0061** (0.0024) | 0.0059** (0.0021) |
| 17 | category_shopping | -0.0023 (0.0059) | 0.0055 (0.0061) | 0.0063 (0.0054) | -0.0118* (0.0052) | -0.0088. (0.0045) | 0.0011 (0.0057) | 0.0002 (0.0059) |
| 18 | category_socializing | 0.0009 (0.0029) | -0.0028 (0.0031) | -0.0025 (0.0029) | 0.0071* (0.0032) | 0.0035 (0.0028) | 0.0012 (0.0031) | 0.0016 (0.0027) |
| 19 | category_eating_or_drinking | 0.0044 (0.0063) | -0.0047 (0.0068) | -0.0039 (0.0058) | 0.0063 (0.0060) | 0.0076 (0.0057) | 0.0043 (0.0061) | -0.0112. (0.0063) |
| 20 | category_playing_sports_or_games | 0.0003 (0.0022) | 0.0031 (0.0025) | -0.0030 (0.0024) | -0.0054. (0.0029) | 0.0008 (0.0024) | -0.0011 (0.0032) | -0.0014 (0.0035) |
| 21 | category_hobby_or_relaxing | 0.0013 (0.0018) | -0.0031 (0.0021) | 0.0032 (0.0021) | 0.0047* (0.0019) | -0.0005 (0.0021) | 0.0027 (0.0021) | 0.0025 (0.0020) |
| 22 | ---------- | ---------- | ---------- | ---------- | ---------- | ---------- | ---------- | ---------- |
| 23 | S.E. type | IID | IID | IID | IID | IID | IID | IID |
| 24 | Observations | 29,615 | 20,143 | 22,851 | 20,442 | 18,980 | 14,036 | 10,930 |
| 25 | Squared Cor. | 0.09618 | 0.08561 | 0.09601 | 0.10968 | 0.09381 | 0.07579 | 0.09919 |
| 26 | Pseudo R2 | 0.09037 | 0.08989 | 0.09449 | 0.10538 | 0.08928 | 0.07543 | 0.08786 |
| 27 | BIC | 22,396.7 | 14,668.1 | 17,965.1 | 15,685.8 | 15,635.8 | 11,663.8 | 9,421.0 |

## A.4 Robustness Checks for Matching

Figure 8: **Robustness Check for Value per Trip Estimates: differing matching equations.** To estimate the value of local recreation, I used nearest neighbor matching to impute the travel time for non-recreators (for whom it is impossible to observe travel time). Here, I show how neither the number of "neighbors" used to impute unobservable travel time nor the matching equation used significantly change the estimated value per trip.

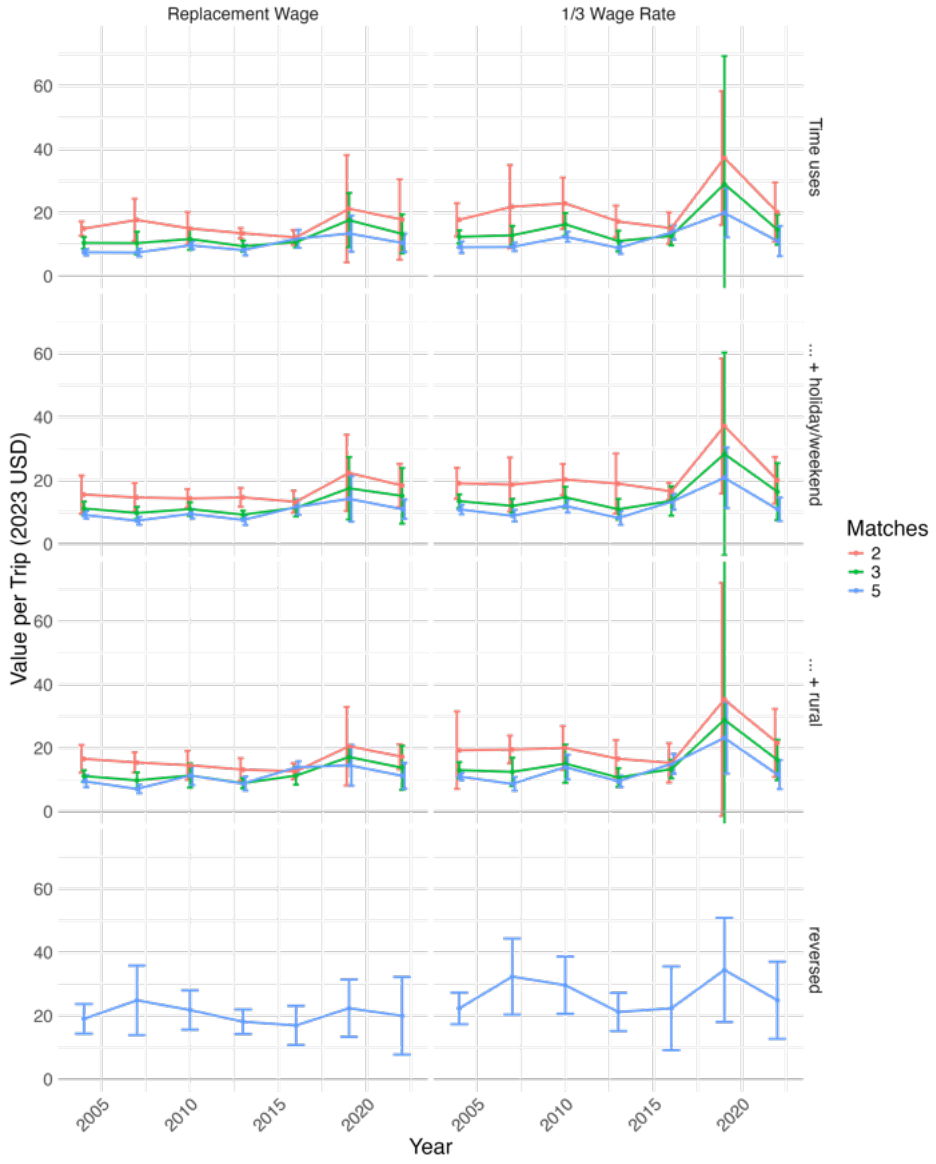


Figure 9: **Robustness Check for Value per Trip, using restricted caliper.**

